Site Frequency Spectrum (SFS)

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Physalia lcWGS course 2026

Learning objectives for today

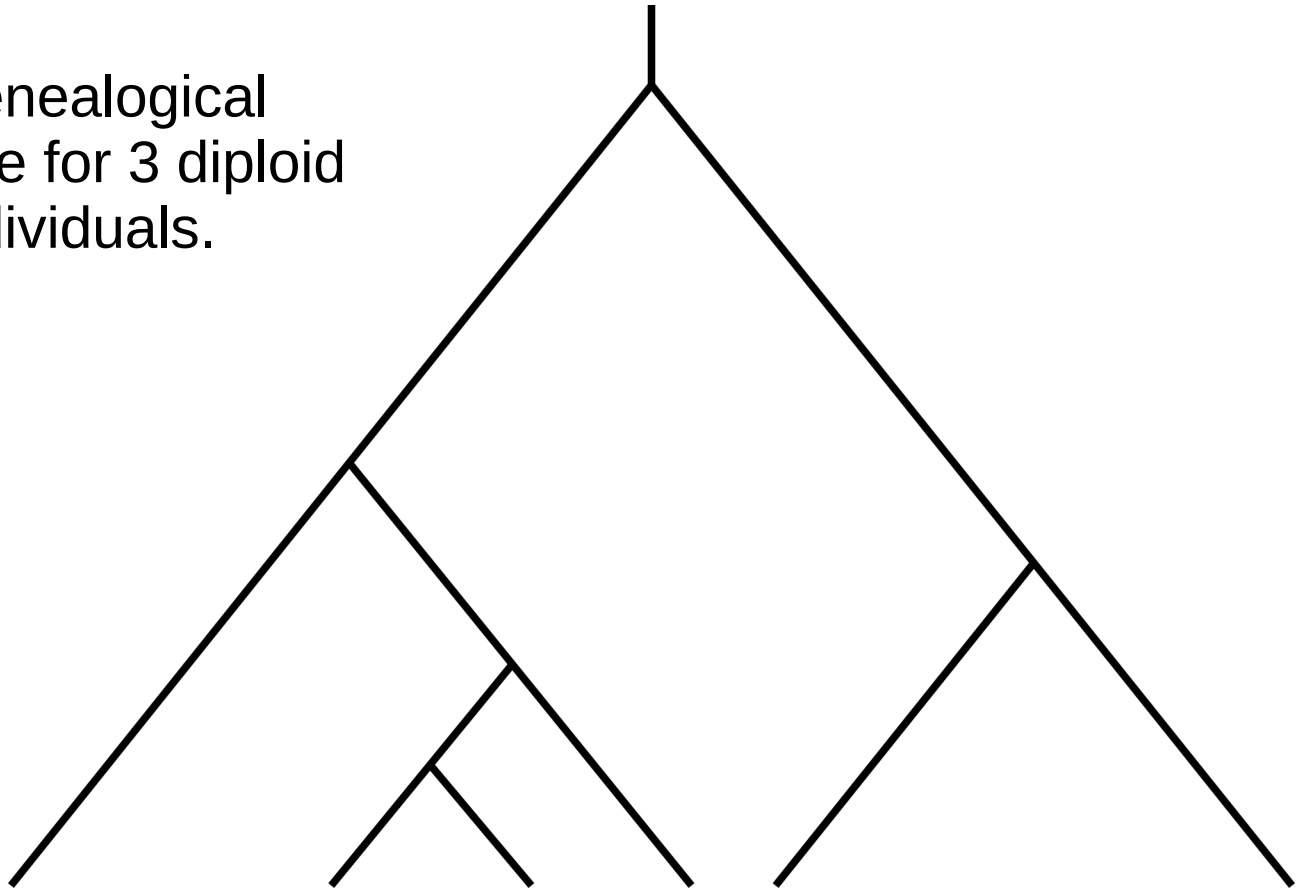
- Understand what the site frequency spectrum is and why it is such an important quantity in population genetic analyses.

- See how the concepts that we have covered in previous days are incorporated into estimating the SFS in a low coverage framework.

- Learn how to estimate the SFS and related quantities with ANGSD and how to interpret the outputs.
    - Individual heterozygosity
    - Thetas (population-level genetic diversity)
    - F_{ST}

- Overview of the types of downstream analyses that you can robustly perform given the quantities that you have learned how to estimate throughout the course.

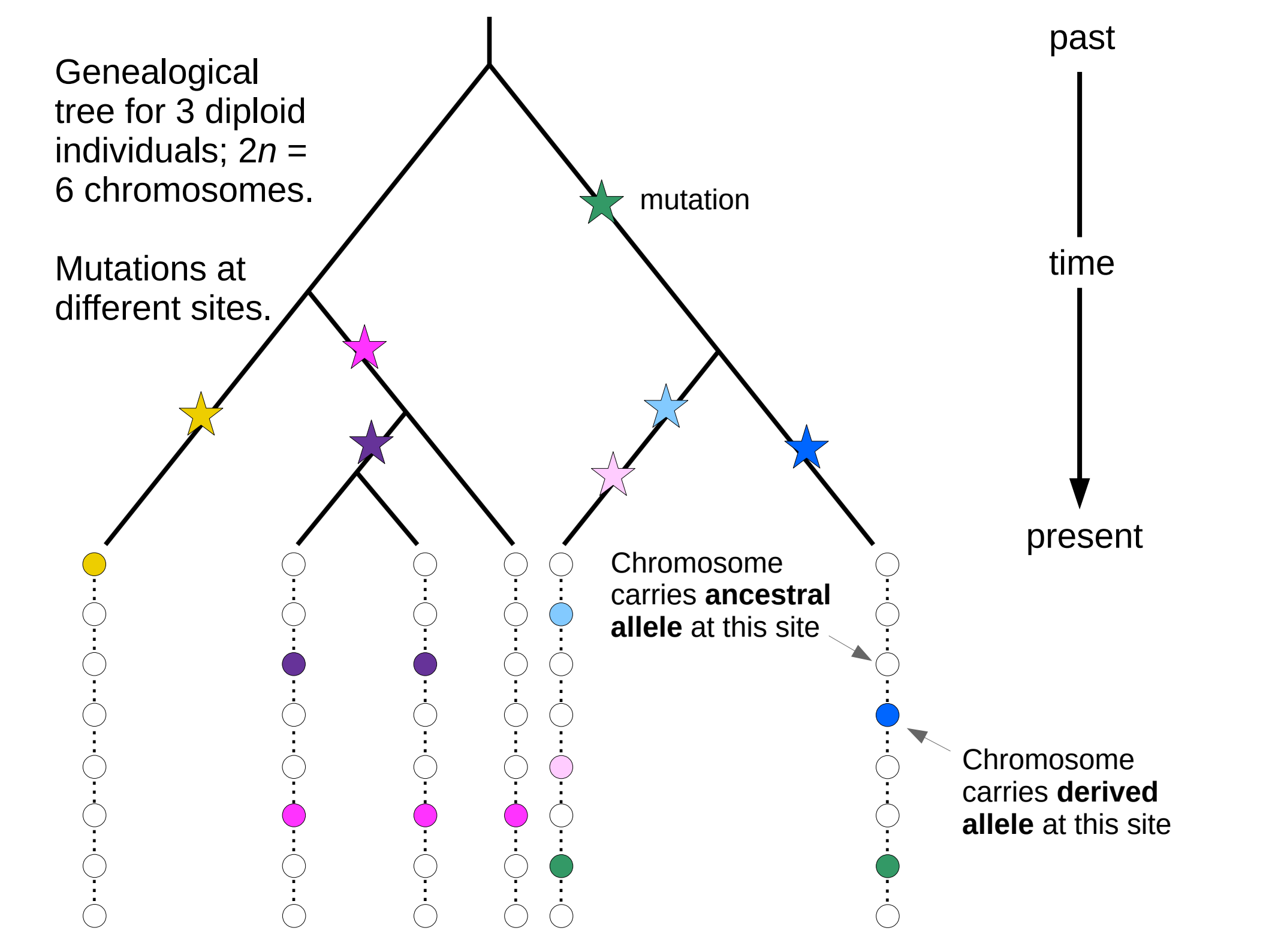
Genealogical tree for 3 diploid individuals.

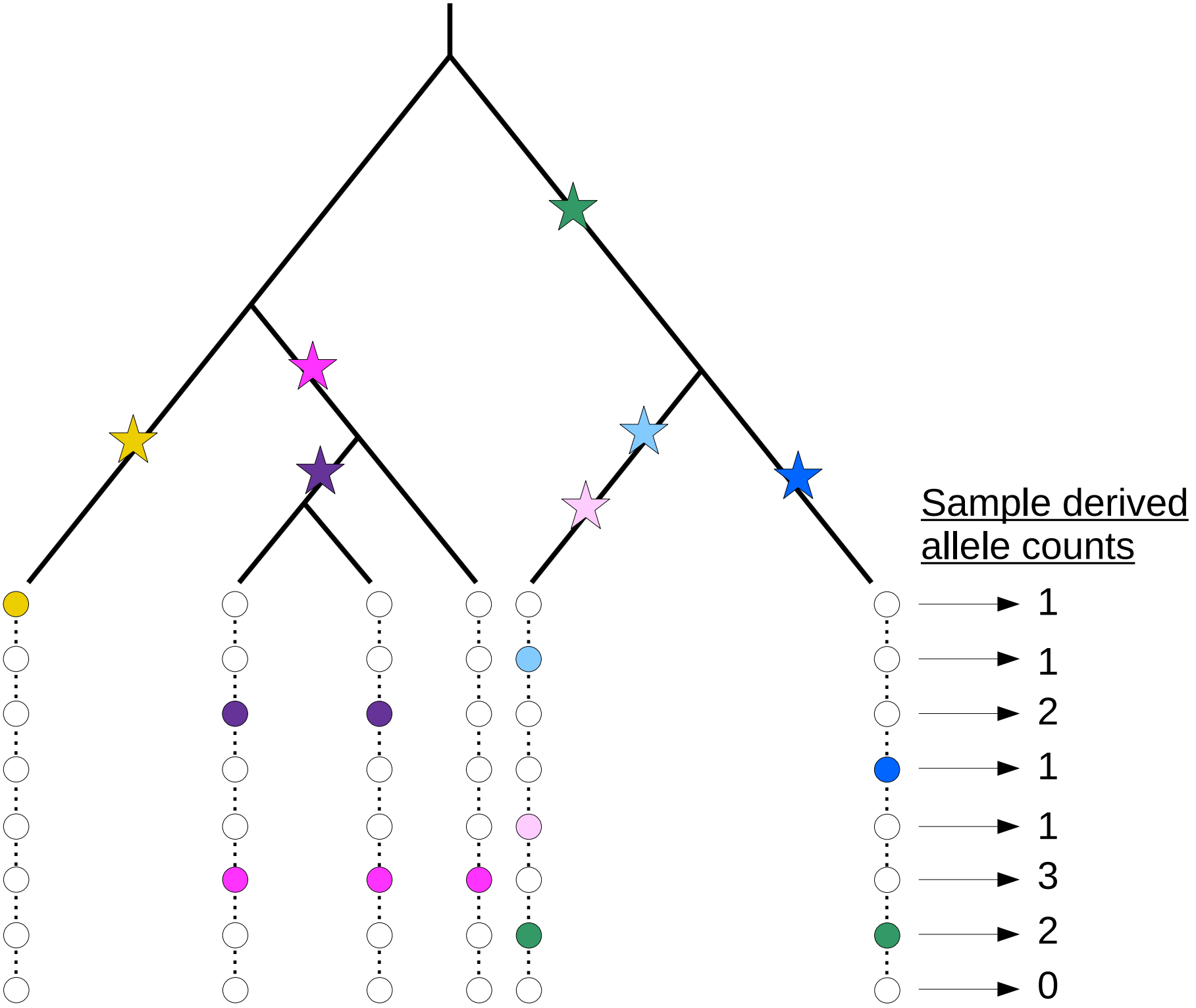


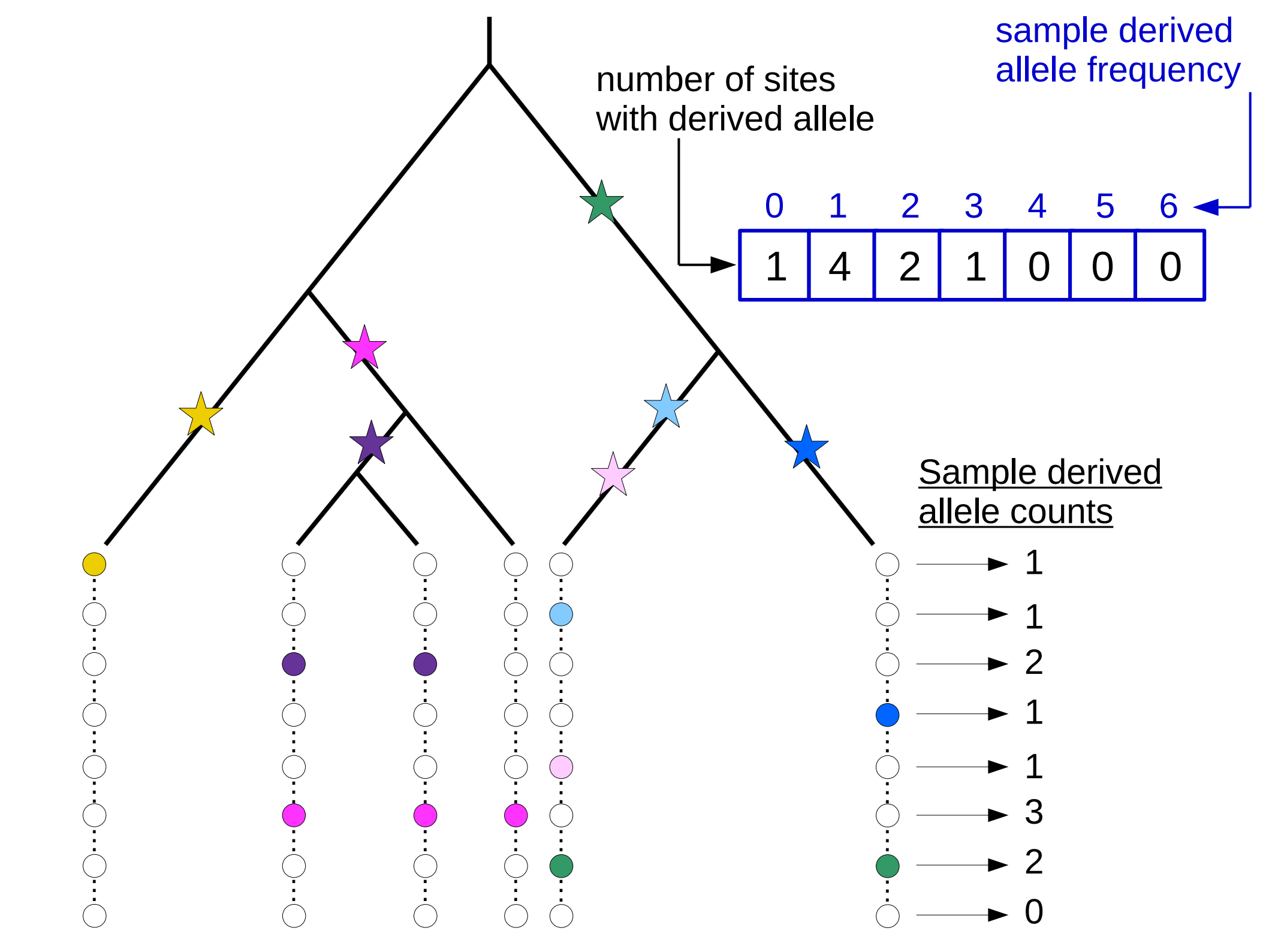
past

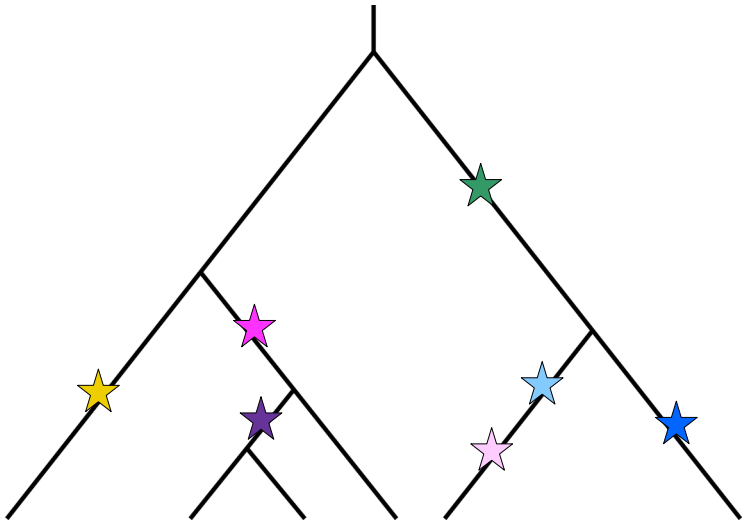
time

present


Genealogical tree for 3 diploid individuals; 2n = 6 chromosomes.
Mutations at different sites.
mutation
past
time
present
Chromosome carries **ancestral allele** at this site
Chromosome carries **derived allele** at this site


Sample derived
allele counts
1
1
2
1
1
3
2
0

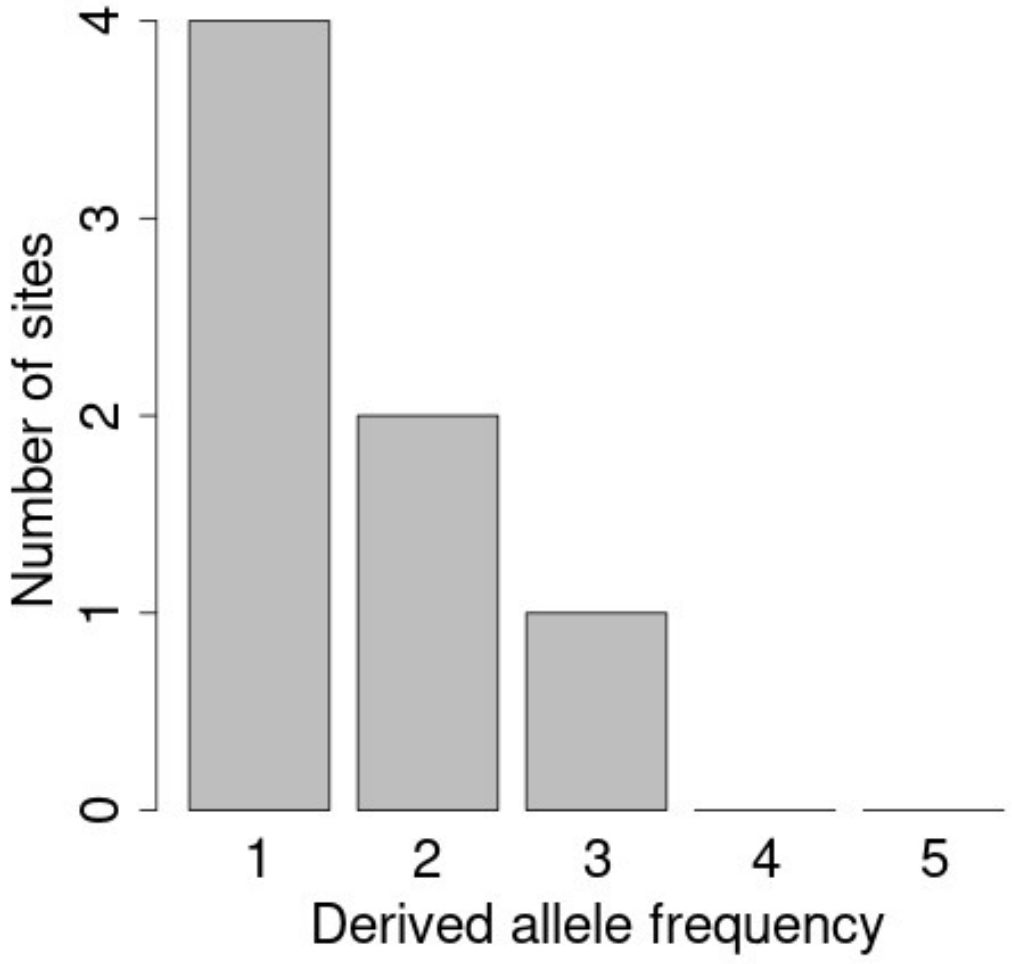

sample derived
allele frequency
number of sites
with derived allele
0 1 2 3 4 5 6
1 4 2 1 0 0 0
Sample derived
allele counts
1
1
2
1
1
3
2
0



number of sites
with derived allele

sample derived
allele frequency

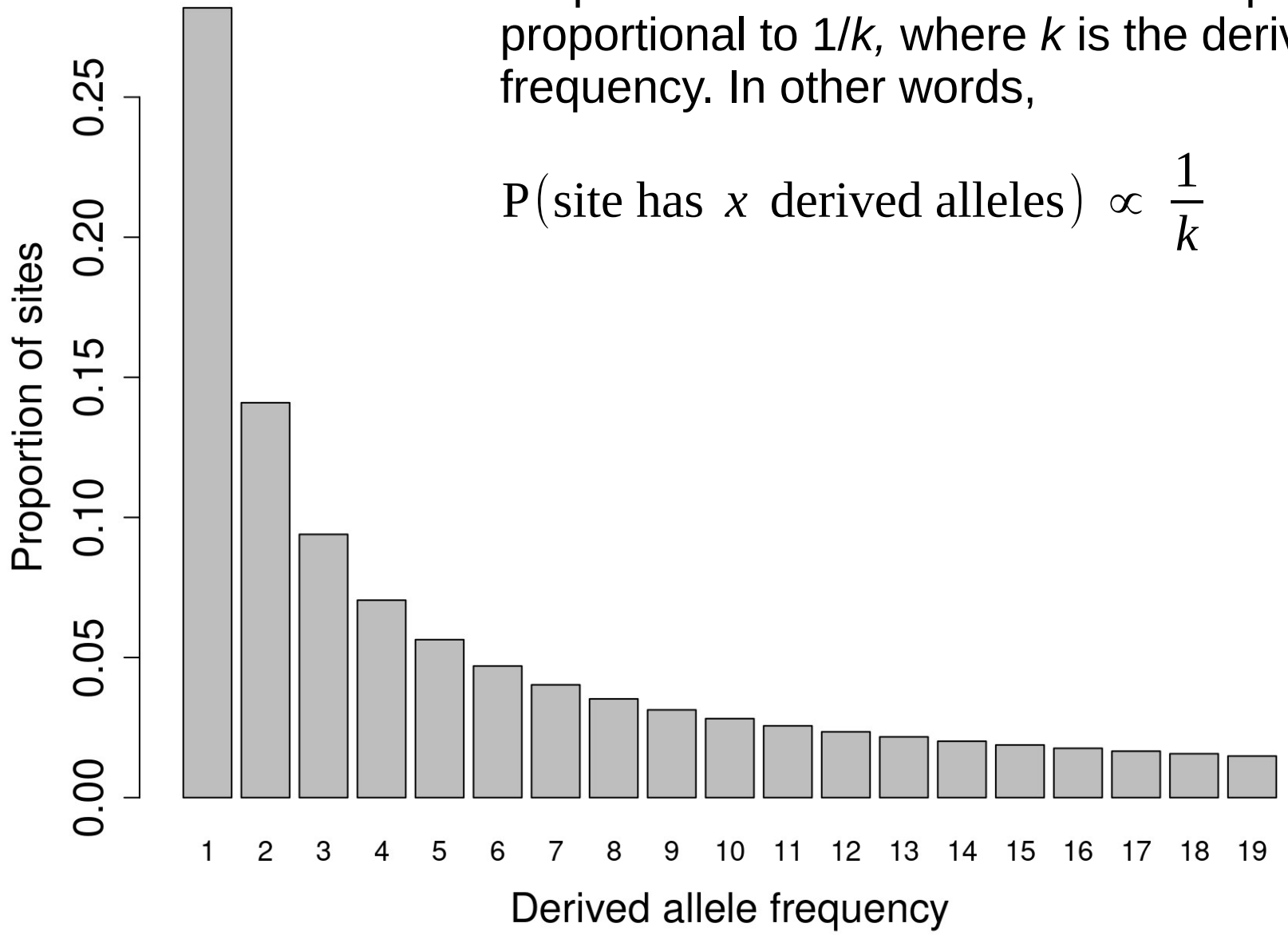
0	1	2	3	4	5	6
1	4	2	1	0	0	0



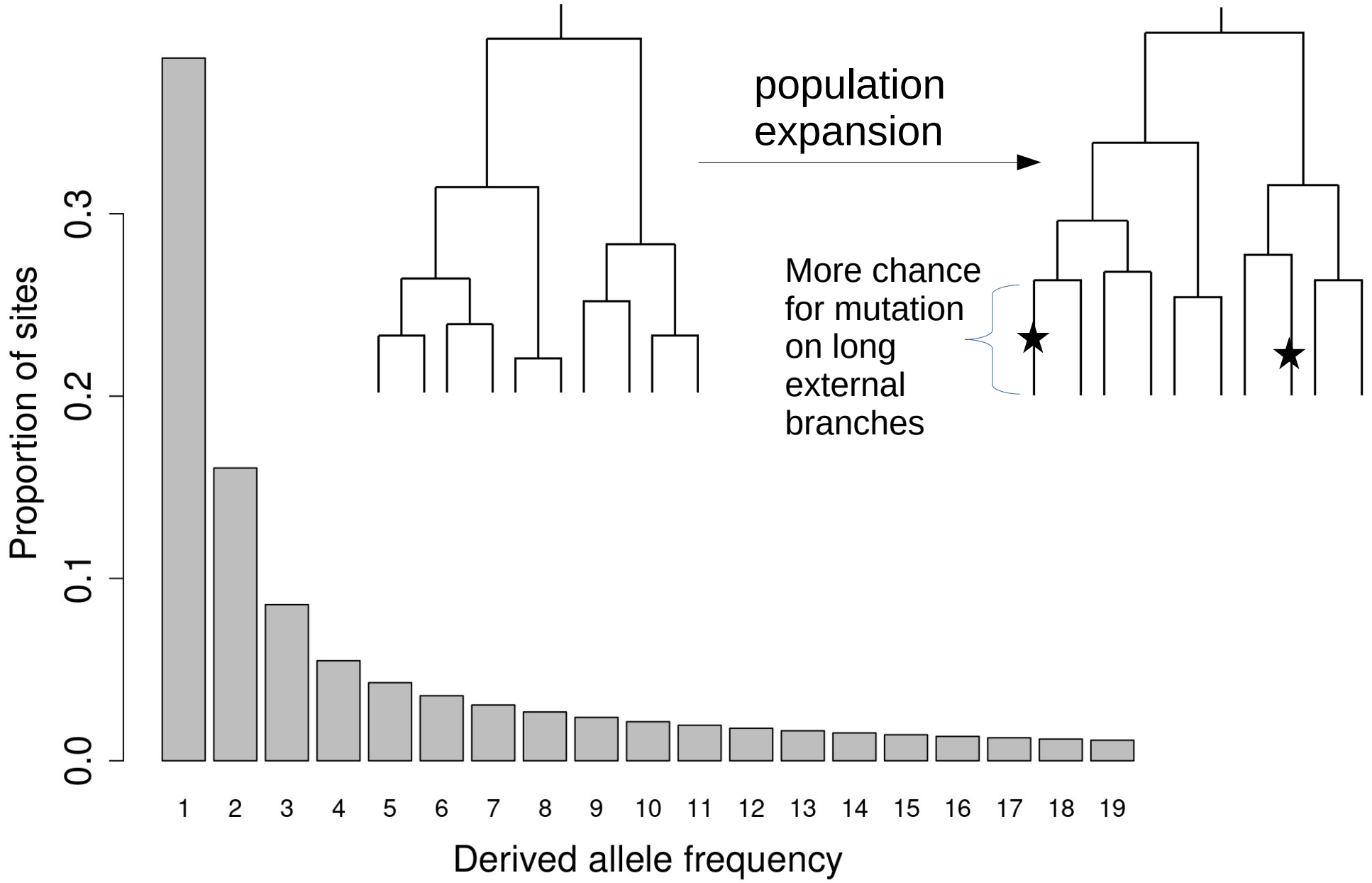
Neutral, constant-size population SFS

Proportion of sites in each allele frequency class proportional to 1/*k,* where *k* is the derived allele frequency. In other words,

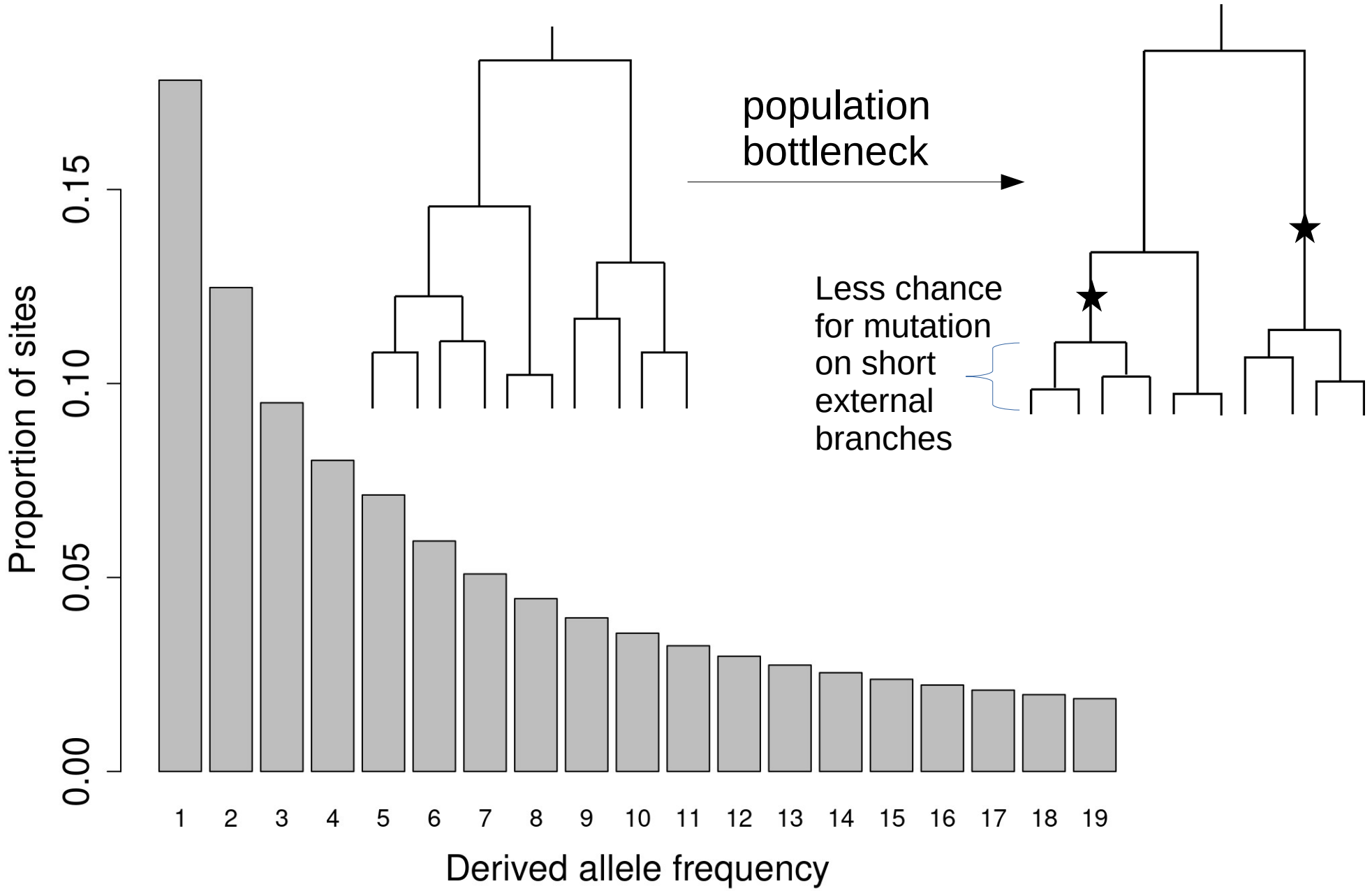
$$\mathrm{P}\left(\text{site has } x \text{ derived alleles}\right) \propto \frac{1}{k}$$



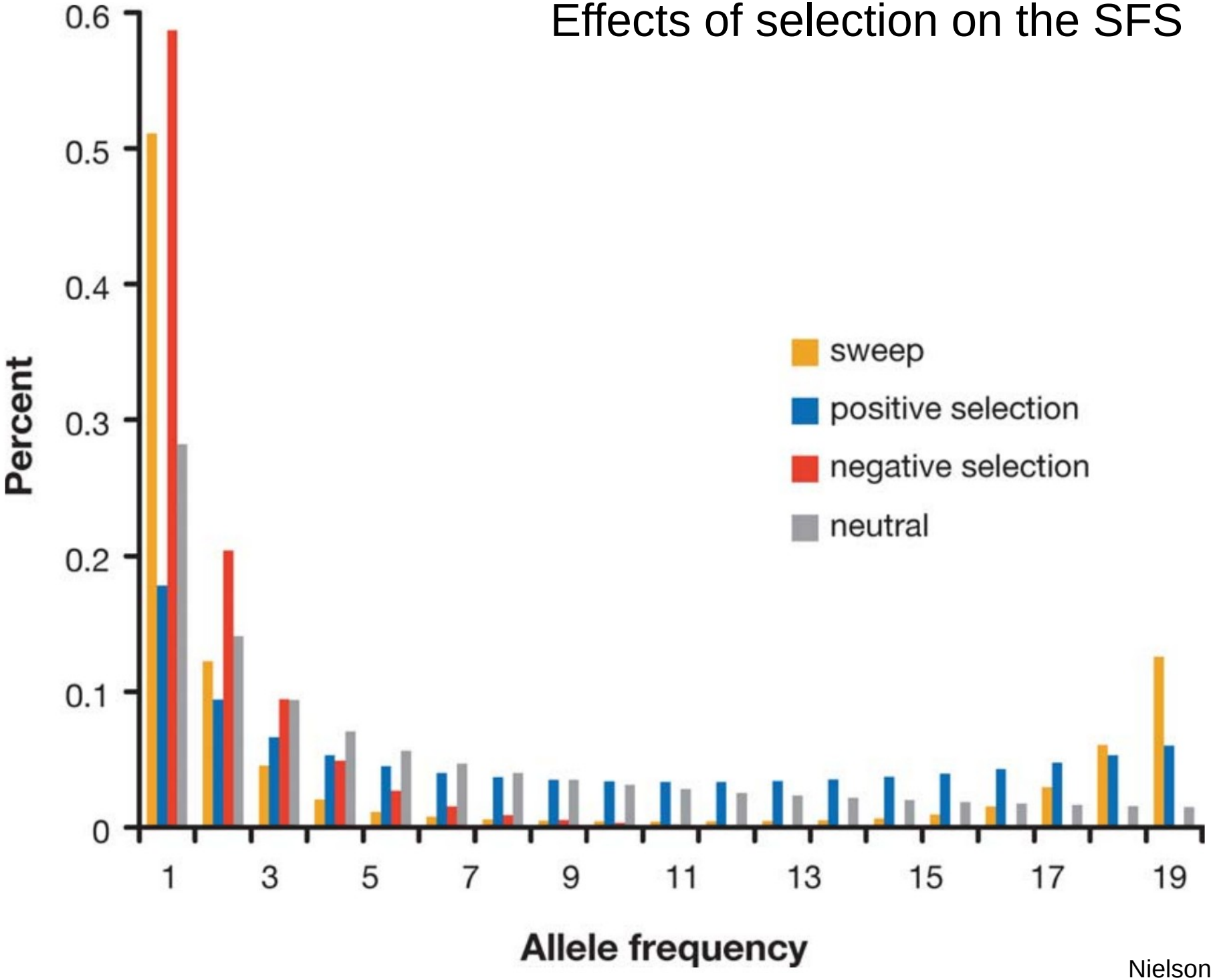
Lineages coalesce at rate 1/*N*. *N* is the effective population size.



Lineages coalesce at rate 1/*N*. *N* is the effective population size.



Effects of selection on the SFS



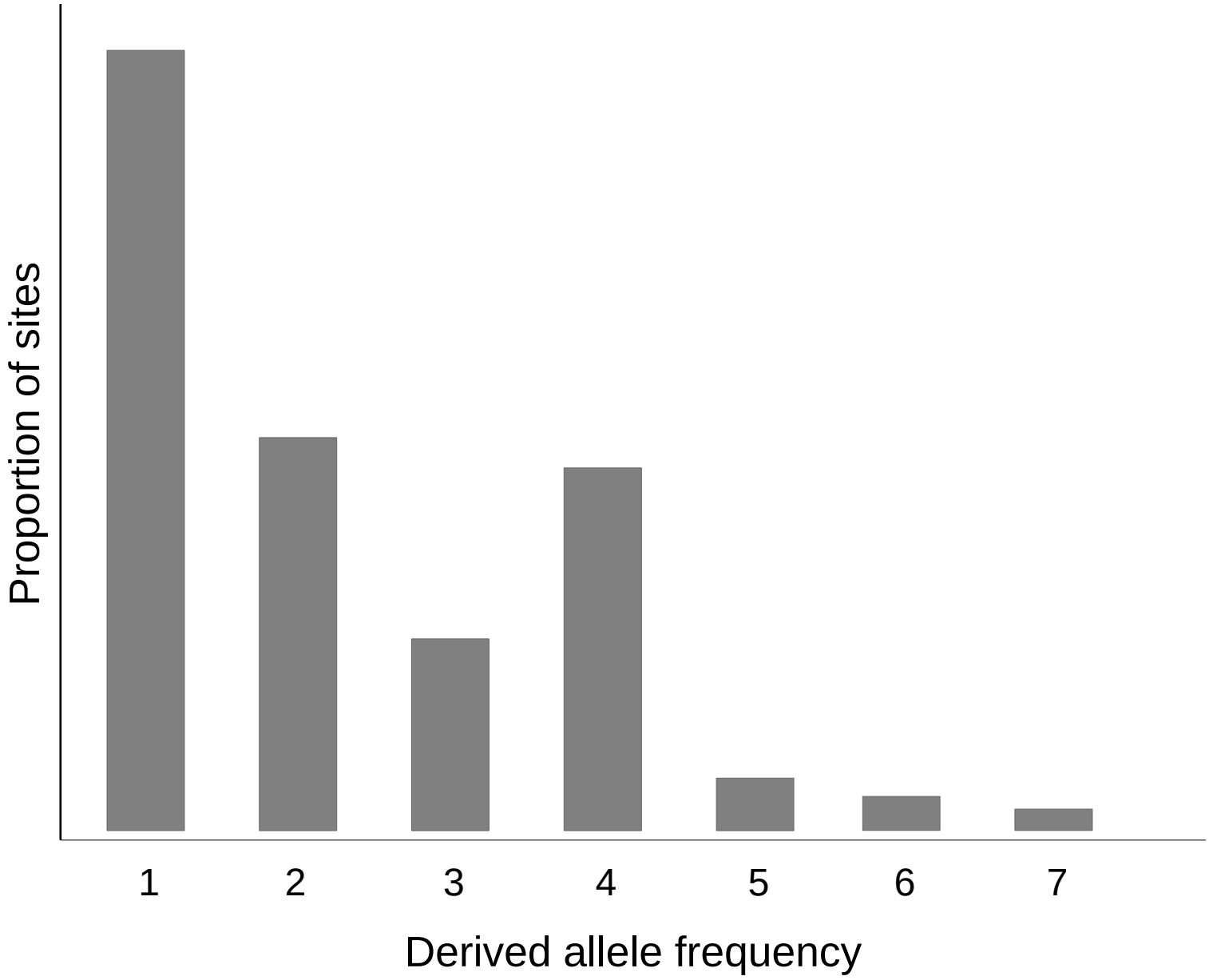
Nielson 2005

Since population genetic theory provides such good expectations for how the SFS should behave, we can use the *genome-wide* SFS to sanity check the reliability of our downstream inferences.

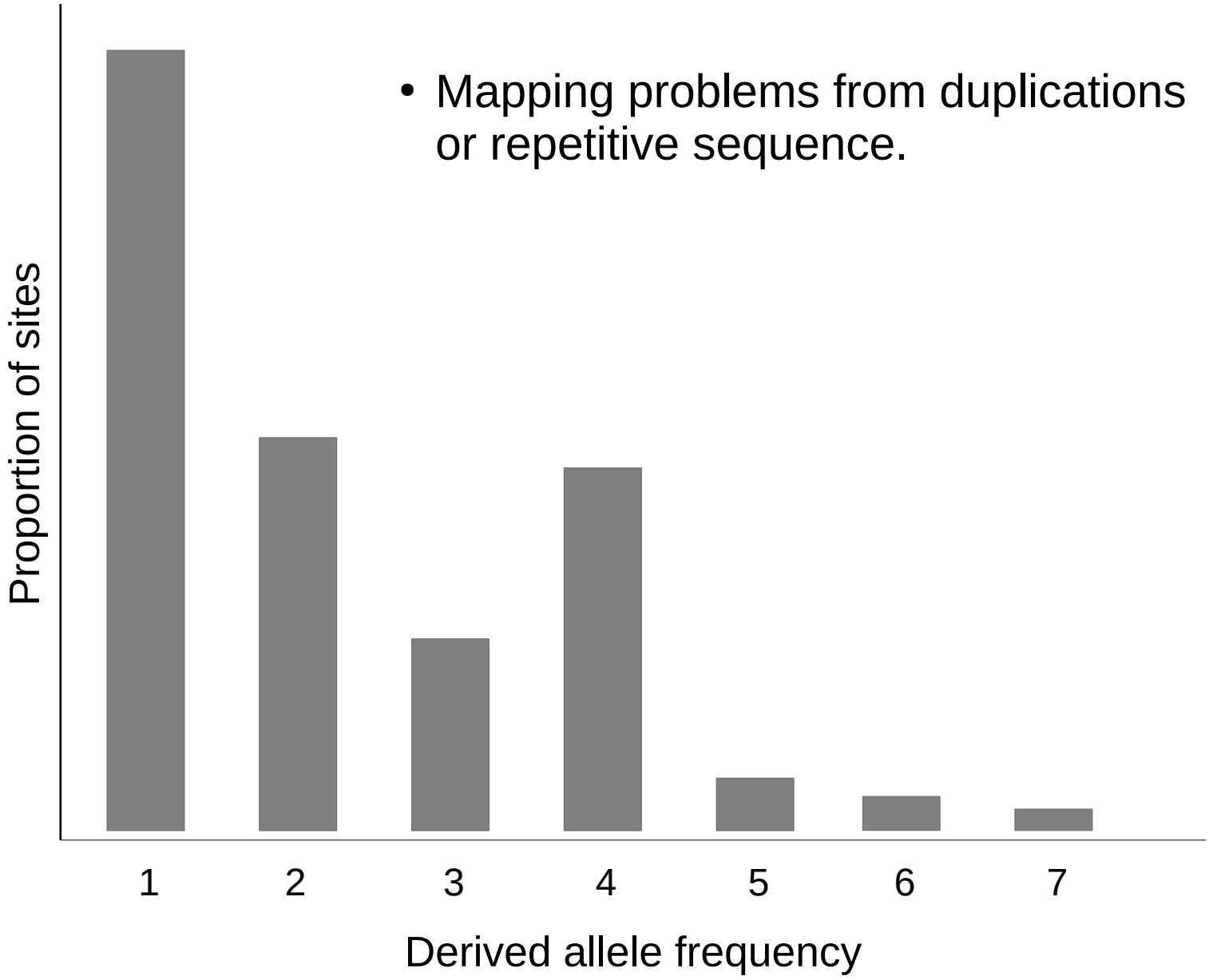
In other words, the SFS can be diagnostic of data quality or processing problems.

Let’s play the guess the bioinformatic problem game

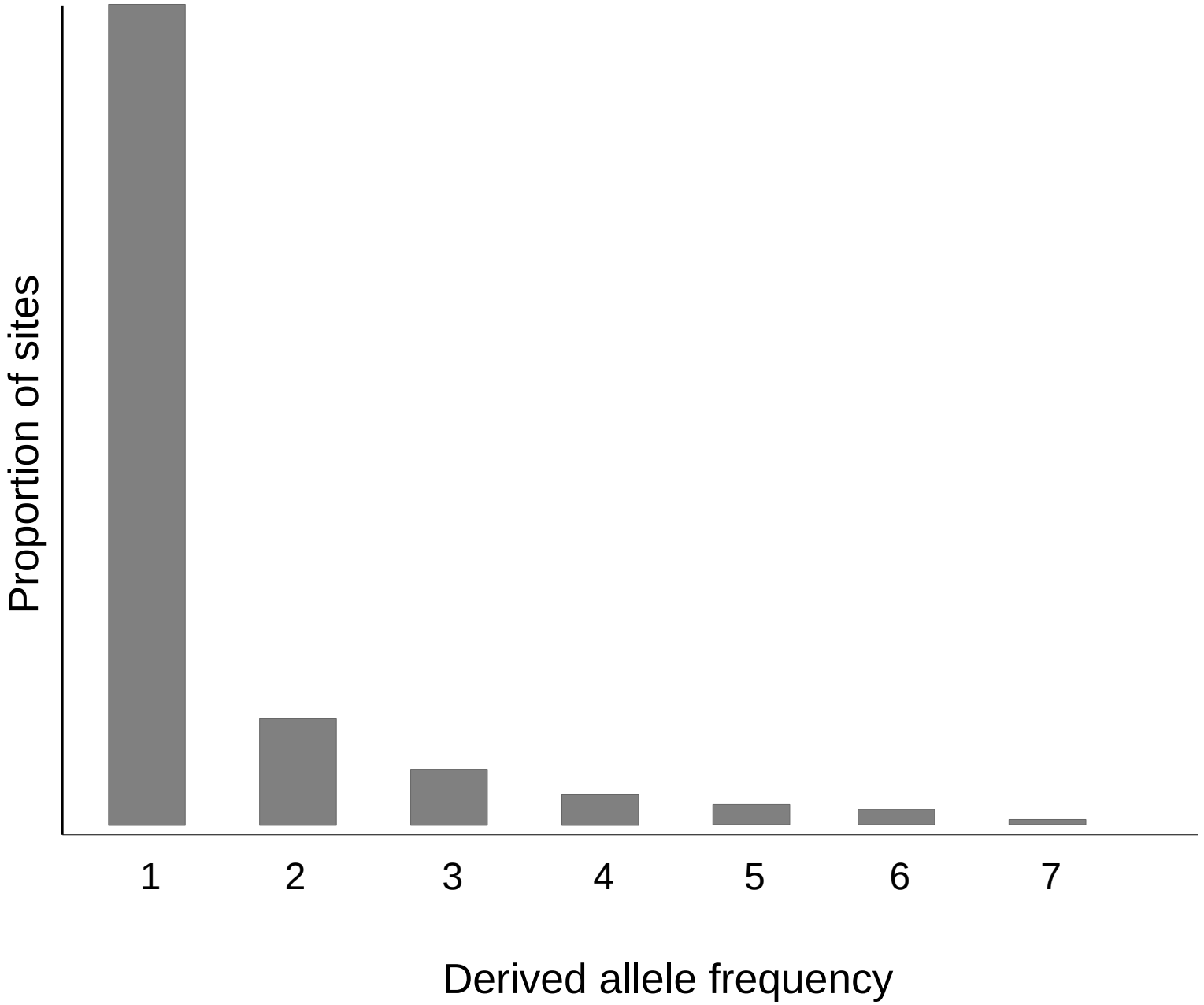
SFS for a sample of 4 diploid individuals
Case 1



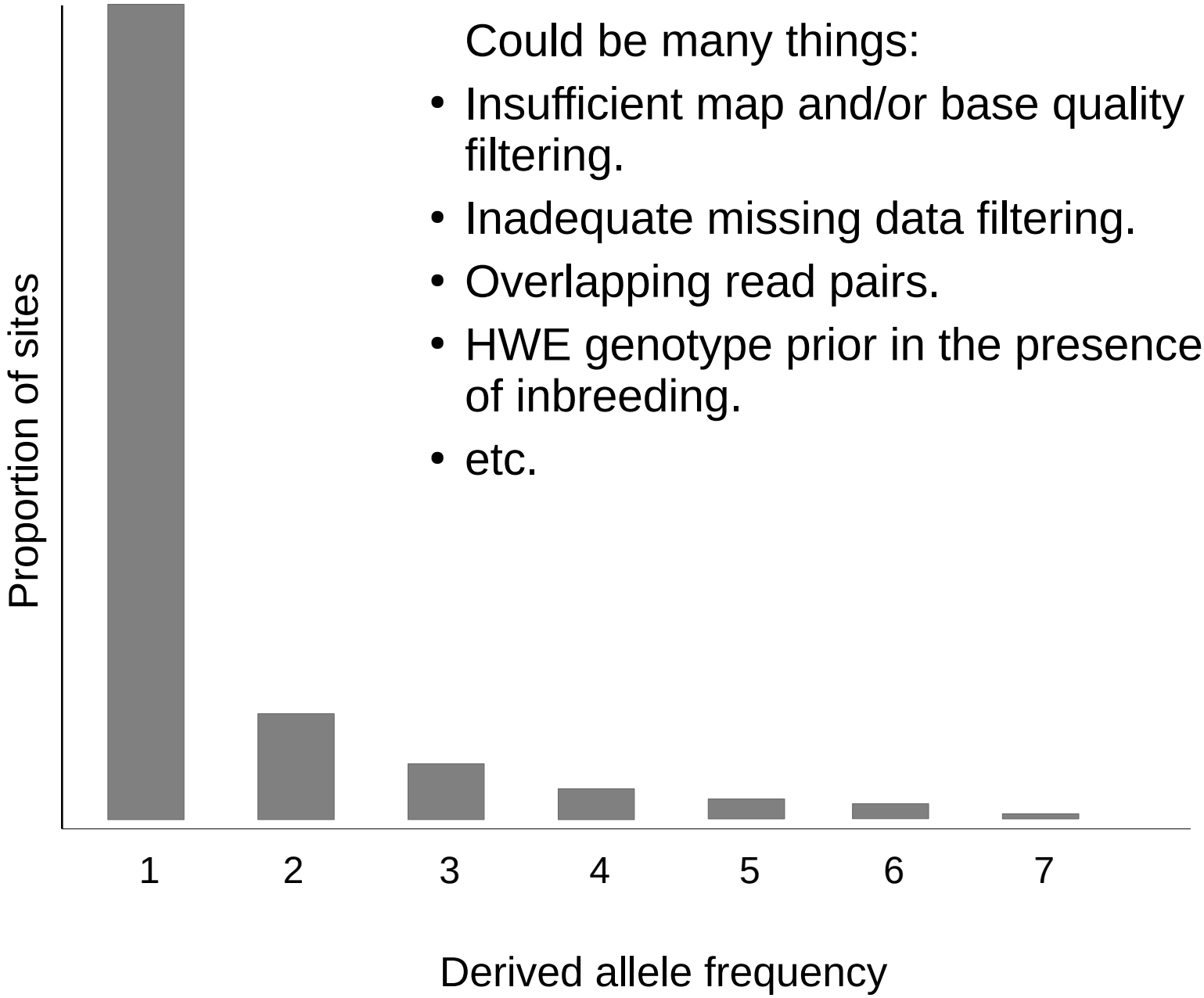
SFS for a sample of 4 diploid individuals
Case 1



SFS for a sample of 4 diploid individuals
Case 2



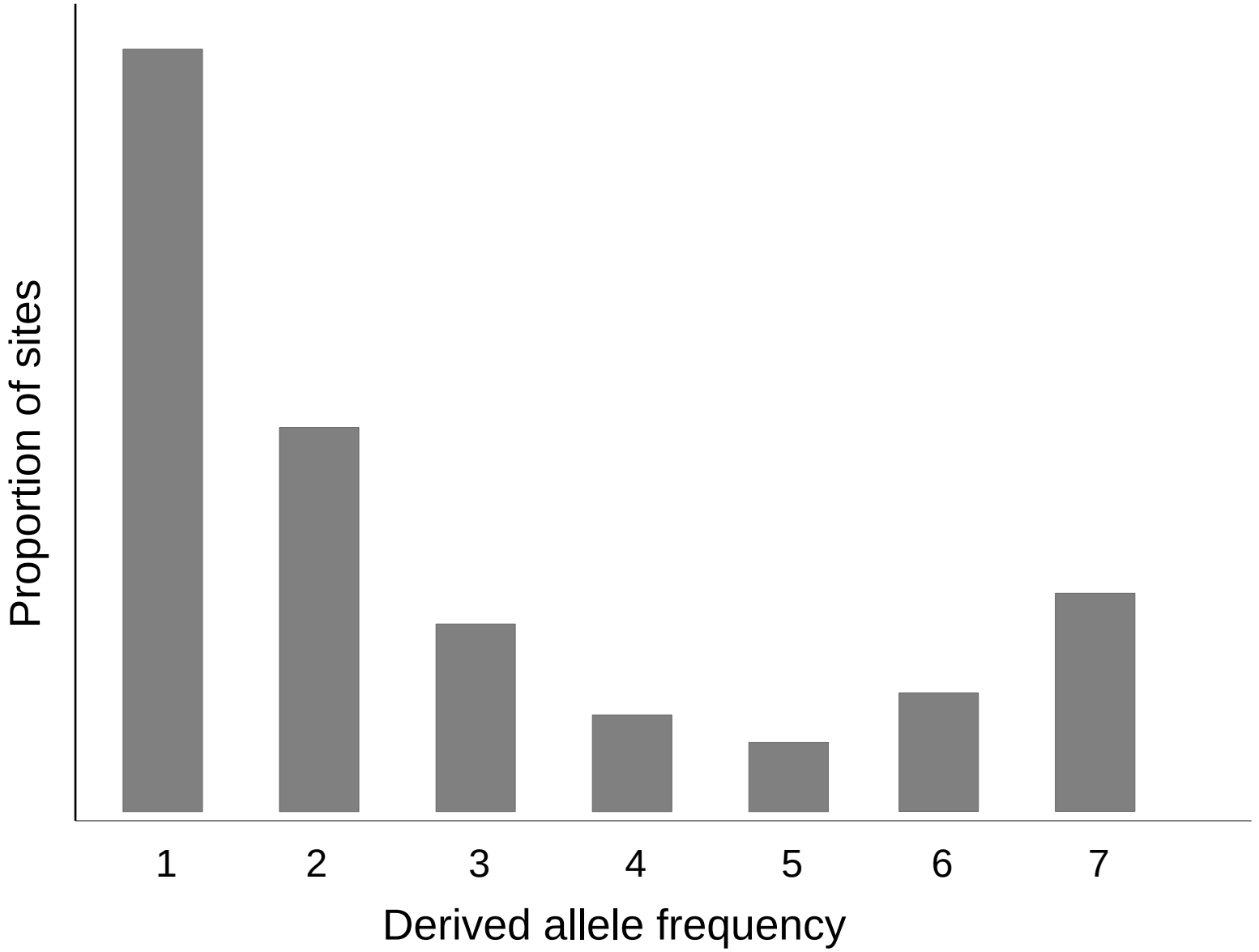
SFS for a sample of 4 diploid individuals
Case 2



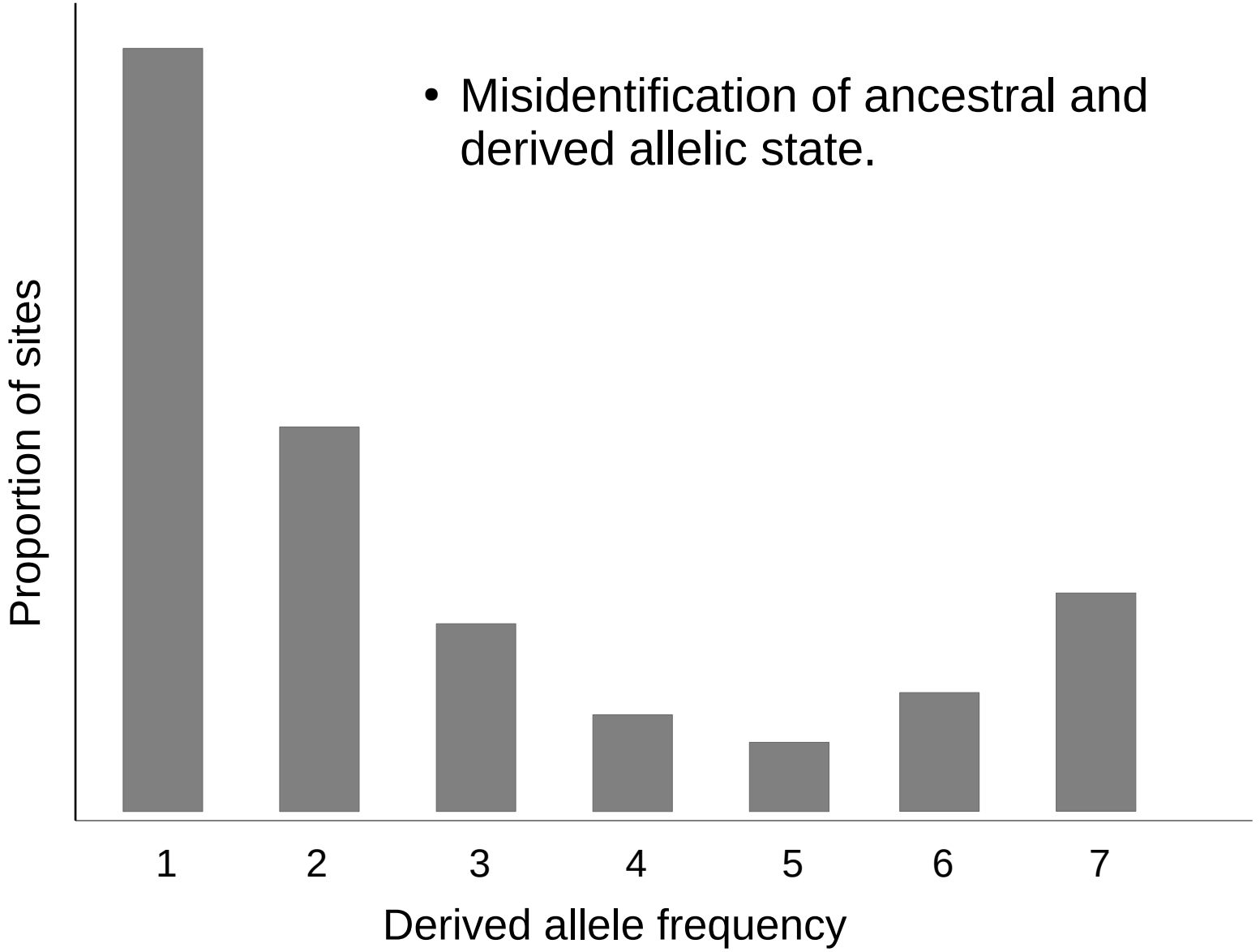
Could be many things:

- Insufficient map and/or base quality filtering.
- Inadequate missing data filtering.
- Overlapping read pairs.
- HWE genotype prior in the presence of inbreeding.
- etc.

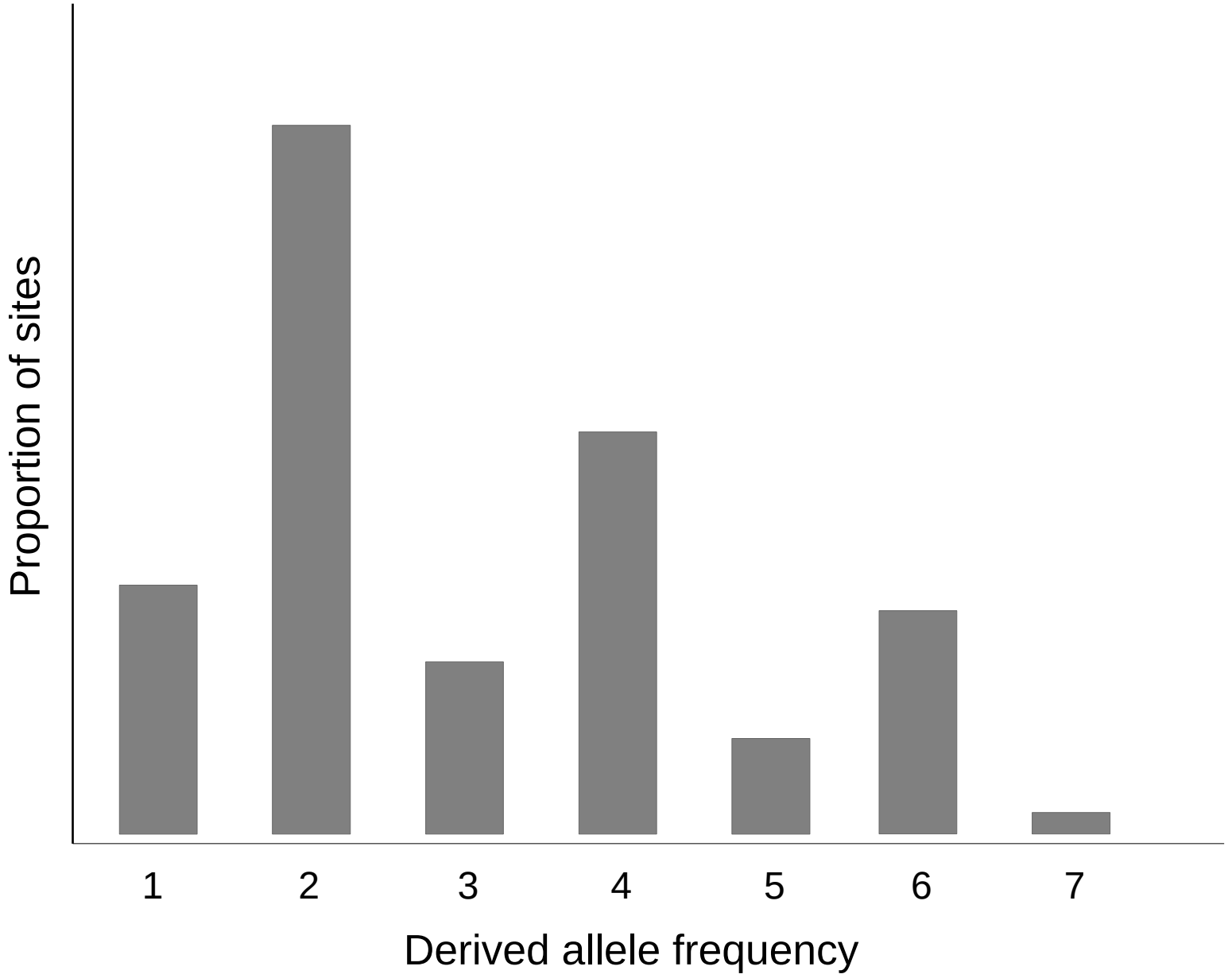
SFS for a sample of 4 diploid individuals
Case 3



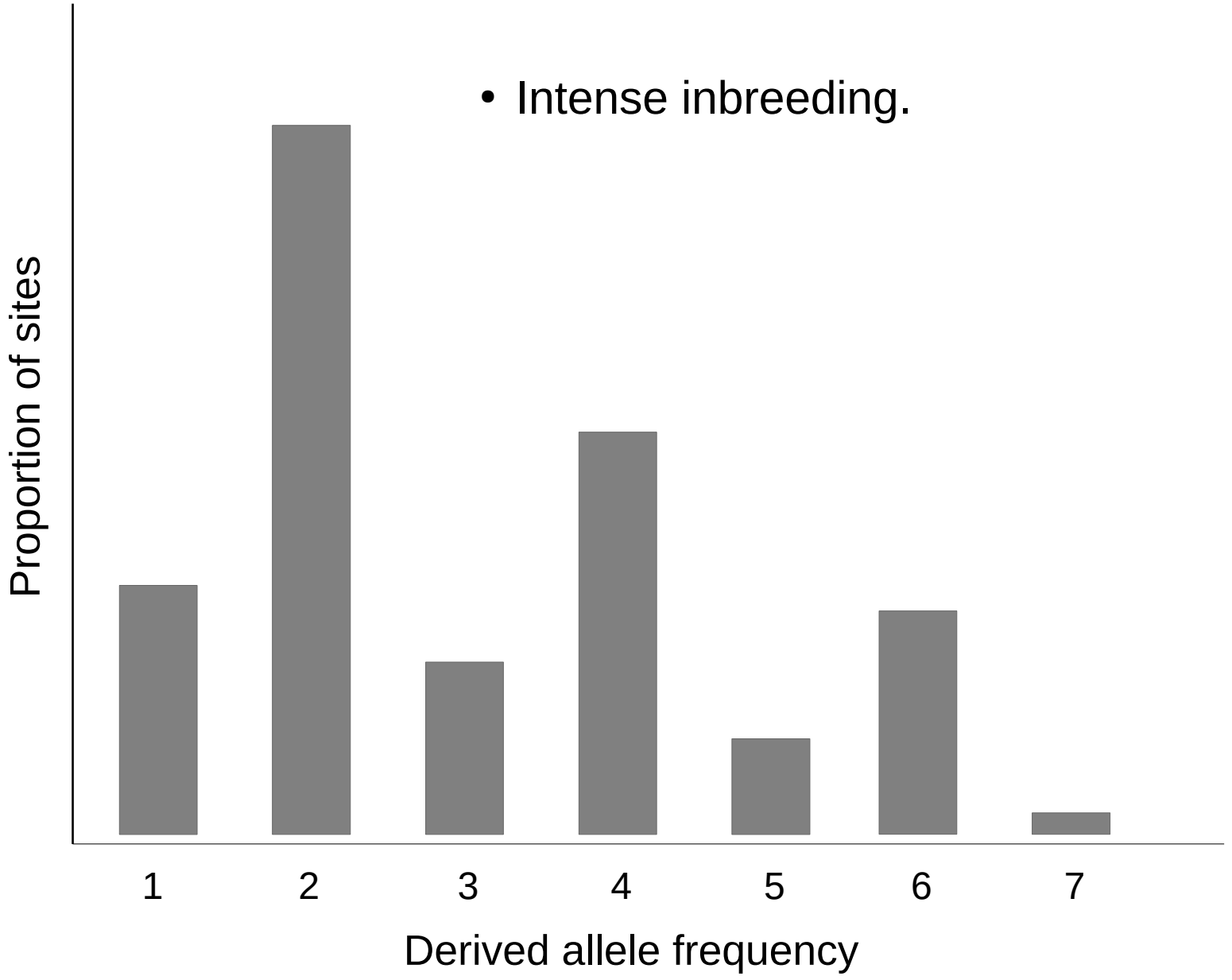
SFS for a sample of 4 diploid individuals
Case 3

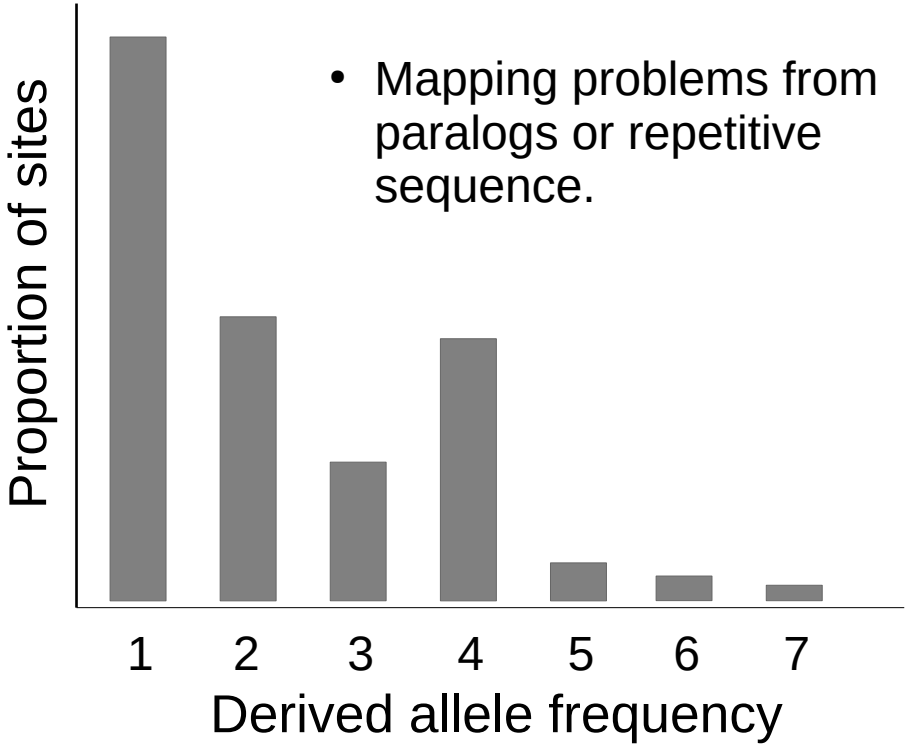


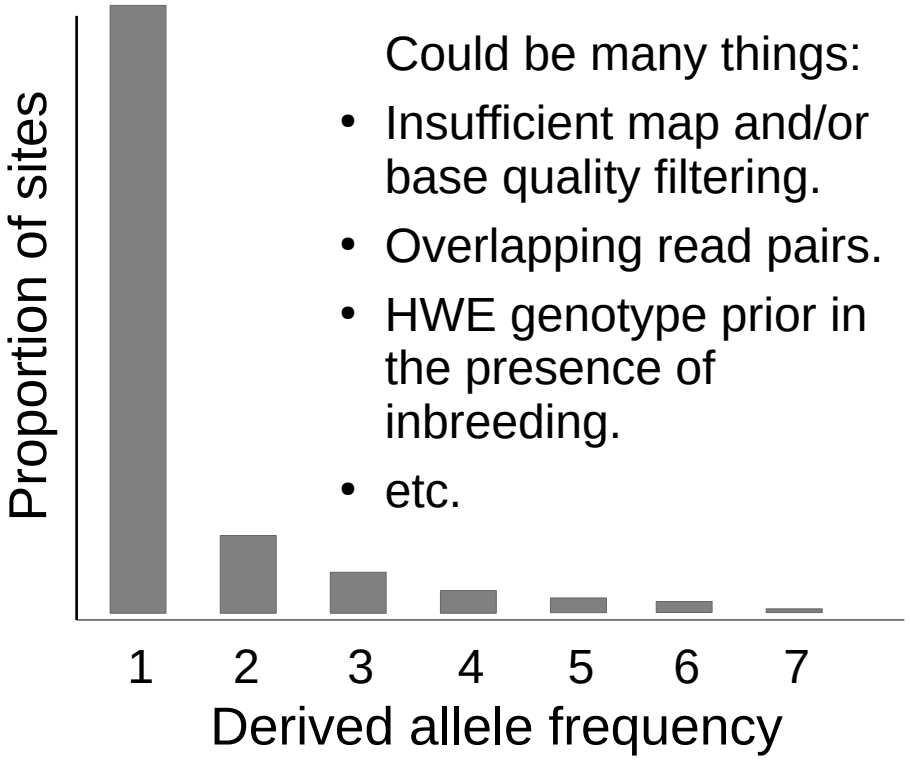
SFS for a sample of 4 diploid individuals
Case 4

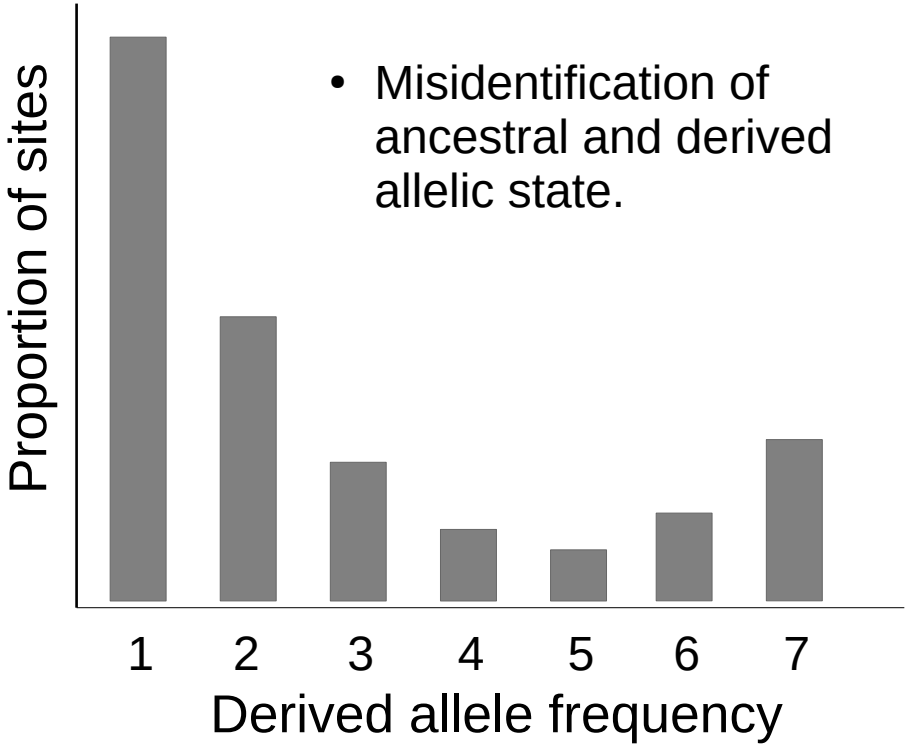


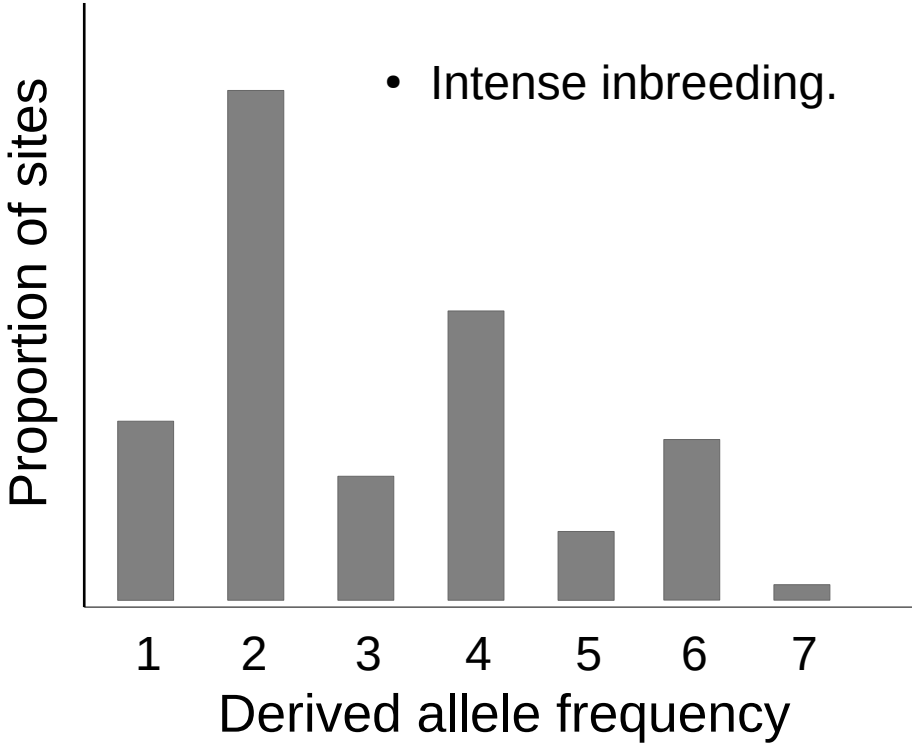
SFS for a sample of 4 diploid individuals
Case 4




Proportion of sites
• Mapping problems from paralogs or repetitive sequence.
1 2 3 4 5 6 7
Derived allele frequency


Proportion of sites
Could be many things:
• Insufficient map and/or base quality filtering.
• Overlapping read pairs.
• HWE genotype prior in the presence of inbreeding.
• etc.
1 2 3 4 5 6 7
Derived allele frequency


Proportion of sites
• Misidentification of ancestral and derived allelic state.
1 2 3 4 5 6 7
Derived allele frequency


Proportion of sites
• Intense inbreeding.
1 2 3 4 5 6 7
Derived allele frequency

Estimating the SFS in a low coverage framework

Likelihood function for the SFS

$$\text{SFS} = \boldsymbol{P} = \left(p_0, p_1, p_2, p_3, p_4, \ldots, p_{2n}\right)$$

p_k: probability that a random site in the genome has k derived alleles

n = diploid sample size

$$\mathrm{SFS} = \boldsymbol{P} = \left(p_0, p_1, p_2, p_3, p_4, \ldots, p_{2n}\right)$$

Likelihood of the SFS with *known* genotypes considering a single site:

X: observed data
(sequencing reads)

$$\mathrm{P}\left(X, \boldsymbol{G} \middle| \boldsymbol{P}\right) = \sum_{k=0}^{2n} \mathrm{P}\left(X, G \middle| K = k\right) \mathrm{P}\left(K = k \middle| \boldsymbol{P}\right)$$

G: genotype vector
= (G_1, G_2, G_3, …, G_n)

K: number of derived alleles in genotype vector ***G***

Likelihood of the SFS with *known* genotypes considering a single site:

$$\mathrm{P}(X, \boldsymbol{G} | \boldsymbol{P}) = \sum_{k=0}^{2n} \mathrm{P}(X, \boldsymbol{G} | K = k) \mathrm{P}(K = k | \boldsymbol{P})$$

$$= p_k$$

Recall that SFS = $\boldsymbol{P} = (p_0, p_1, \ldots, p_{2n})$

$$\mathrm{P}(a, b | c) = \mathrm{P}(a | b, c) \mathrm{P}(b | c)$$

Noting that the observed data only depends on the genotypes

$$= \sum_{k=0}^{2n} \mathrm{P}(X | \boldsymbol{G}) \mathrm{P}(\boldsymbol{G} | K = k) \mathrm{P}(K = k | \boldsymbol{P})$$

Likelihood of the SFS with *known* genotypes considering a single site:

$$\mathrm{P}(X, \boldsymbol{G} | \boldsymbol{P}) = \sum_{k=0}^{2n} \mathrm{P}(X, \boldsymbol{G} | K = k) \mathrm{P}(K = k | \boldsymbol{P})$$

$$\mathrm{P}(a, b | c) = \mathrm{P}(a | b, c) \mathrm{P}(b | c)$$

$$= p_k$$

Recall that SFS = $\boldsymbol{P} = (p_0, p_1, \ldots, p_{2n})$

$$= \sum_{k=0}^{2n} \mathrm{P}(X | \boldsymbol{G}) \mathrm{P}(\boldsymbol{G} | K = k) \mathrm{P}(K = k | \boldsymbol{P})$$

Consider each individual's data and genotype.

$$= \sum_{k=0}^{2n} \left(\prod_{i=1}^{n} \mathrm{P}(X_i | G_i) \right) \mathrm{P}(\boldsymbol{G} | K = k) \mathrm{P}(K = k | \boldsymbol{P})$$

Likelihood of the SFS with *known* genotypes considering a single site:

$$\mathrm{P}(X, \boldsymbol{G}|\boldsymbol{P}) = \sum_{k=0}^{2n} \left(\prod_{i=1}^{n} \mathrm{P}(X_i|G_i) \right) p(\boldsymbol{G}|K=k)\, \mathrm{P}(K=k|\boldsymbol{P})$$

Probability based on the number of ways to have *k* derived alleles in ***G*** out of the total number of ways to to arrange *k* derived alleles among 2*n* sampled chromosomes. ⟶ **Assumes HWE**

Example for 4 diploid individual with derived allele *A* and ancestral allele *a*.

k = 4

m = # heterozygotes = 2

AA	aA	aa	aA
AA	Aa	aa	aA
AA	aA	aa	Aa
AA	Aa	aa	Aa

$2^m = 2^2$
= 4 combinations

$$\mathrm{P}(\boldsymbol{G}|K=k) = \frac{2^m}{\binom{2n}{k}} = \frac{2^2}{\binom{8}{4}} = \frac{4}{70}$$

$$\mathrm{P}(X, \boldsymbol{G}|\boldsymbol{P}) = \sum_{k=0}^{2n} \left(\prod_{i=1}^{n} \mathrm{P}(X_i|G_i) \right) p(\boldsymbol{G}|K=k)\,\mathrm{P}(K=k|\boldsymbol{P})$$

Allow for unknown genotypes (consider the likelihood for all possible genotypes)

$$\mathrm{P}(X|\boldsymbol{P}) =$$

$$\sum_{k=0}^{2n} \mathrm{P}(K=k|\boldsymbol{P}) \sum_{G_1 \in \{0,1,2\}} \cdots \sum_{G_n \in \{0,1,2\}} \mathrm{P}(G|K=k) \prod_{i=1}^{n} \mathrm{P}(X_i|G_i)$$

Summing over the uncertainty of the identity of genotypes contained in **G**

Likelihood of **P** considering site *v*

$$\mathrm{P}\left(X^v|\boldsymbol{P}\right)=$$

$$\sum_{k=0}^{2n}\mathrm{P}\left(K=k|\boldsymbol{P}\right)\sum_{G_1\in\{0,1,2\}}\cdots\sum_{G_i\in\{0,1,2\}}\mathrm{P}\left(G^v|K=k\right)\prod_{i=1}^{n}\mathrm{P}\left(X_i^v|G_i^v\right)$$

Assume sites are independent

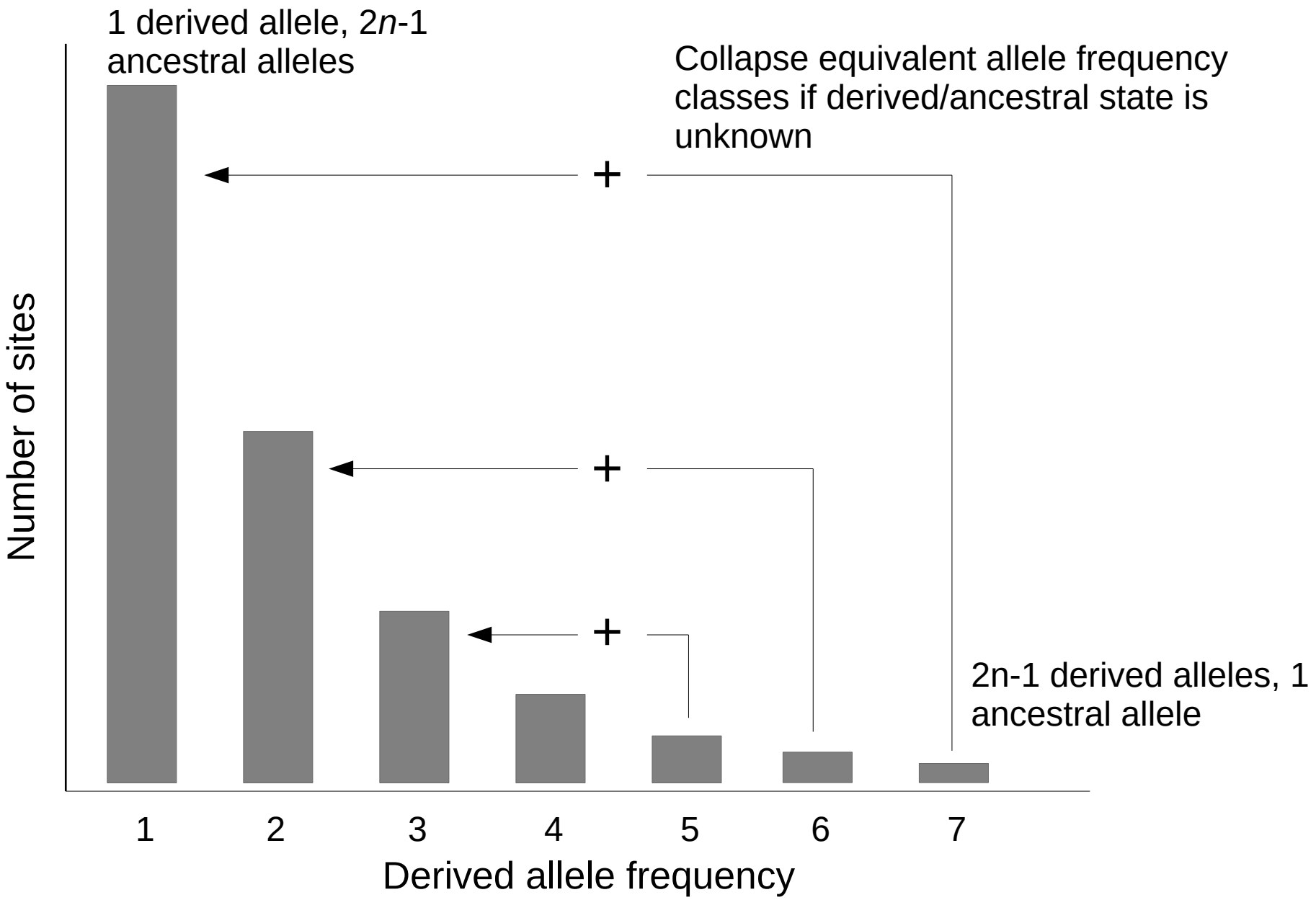
Likelihood of **P** considering all sites (the SFS)

$$\mathrm{P}\left(X|\boldsymbol{P}\right)=$$

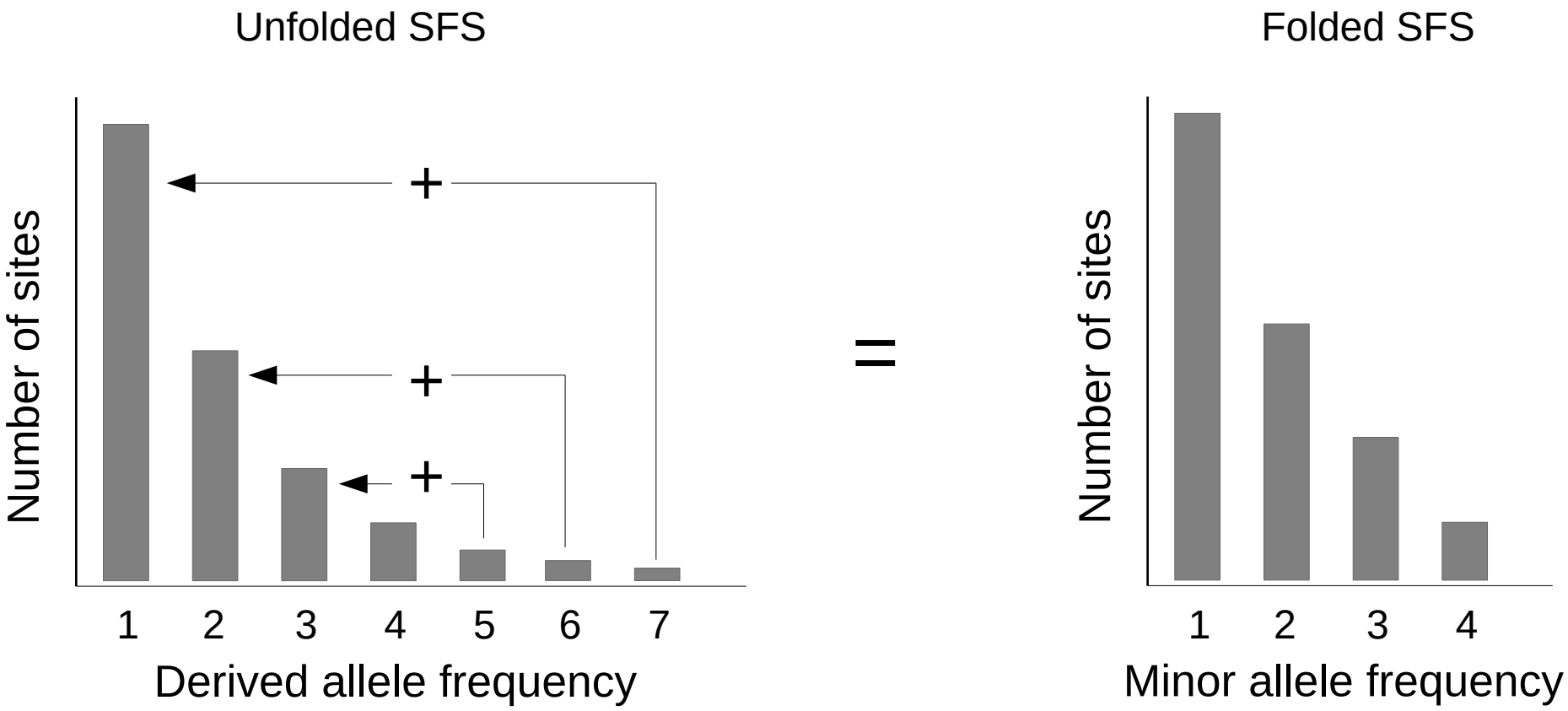
$$\prod_{v=1}^{\text{all sites}}\sum_{k=0}^{2n}\mathrm{P}\left(K=k|\boldsymbol{P}\right)\sum_{G_1\in\{0,1,2\}}\cdots\sum_{G_d\in\{0,1,2\}}\mathrm{P}\left(G^v|K=k\right)\prod_{d=1}^{n}\mathrm{P}\left(X_i^v|G_i^v\right)$$

Product over all sites

Folding the SFS when ancestral state is unknown



Folding the SFS when ancestral state is unknown



ANGSD -doSaf 1 calculates a table of allele frequency likelihoods like the one below.

Use these likelihoods with the realSFS subprogram to obtain a maximum likelihood estimate of the genome-wide SFS.

	0	**1**	**2**	**3**	**4**	**.**	**.**	**.**	***2n*** ←	# derived alleles
site1	0.00	-2.24	-4.53	-6.99	-9.63				-232.69	
site2	0.00	-2.24	-4.53	-6.99	-9.63				-232.69	
site3	-76.63	-37.87	-10.42	0.00	-9.59				-467.13	
site4	0.00	-2.24	-5.53	-6.99	-9.63				-237.55	
.										
.										
.										
.										
.										
.										
site*m*	**0.00**	**-8.62**	**-19.22**	**-30.67**	**-43.27**				**-626.78**	

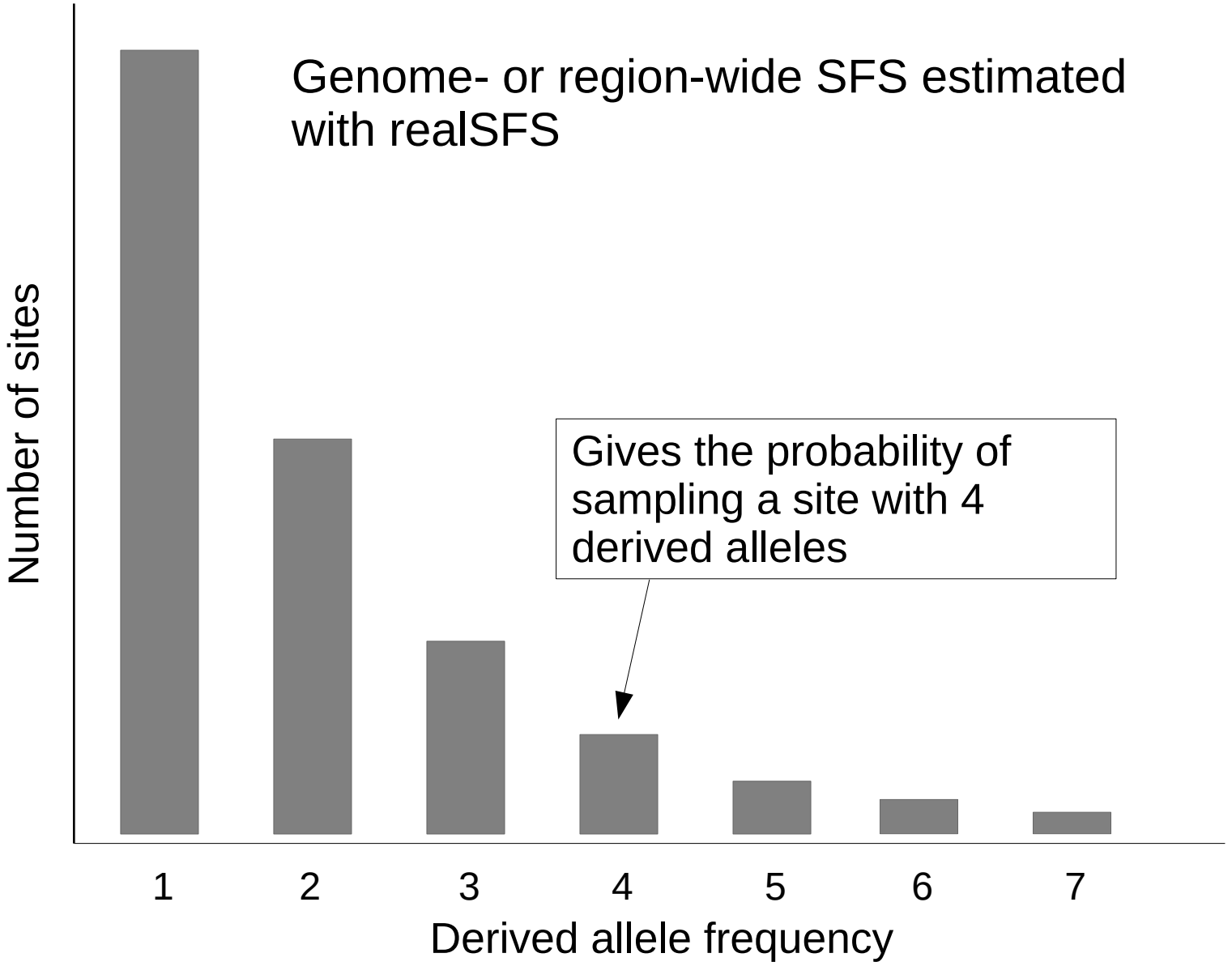
How to estimate posterior probabilities of allele frequencies: allele frequency likelihoods

ANGSD -doSaf 1: Table of allele frequency likelihoods

	0	**1**	**2**	**3**	**4**	.	.	.	***2n*** ◄	# derived alleles
site1	0.00	-2.24	-4.53	-6.99	-9.63				-232.69	
site2	0.00	-2.24	-4.53	-6.99	-9.63				-232.69	
site3	-76.63	-37.87	-10.42	0.00	-9.59				-467.13	
site4	0.00	-2.24	-5.53	-6.99	-9.63				-237.55	
.										
.										
.										
.										
.										
.										
site*m*	**0.00**	**-8.62**	**-19.22**	**-30.67**	**-43.27**				**-626.78**	

$$P(\theta|X) = P(X|\theta)\,P(\theta)$$

How to estimate posterior probabilities of allele frequencies: Allele frequency prior probabilities



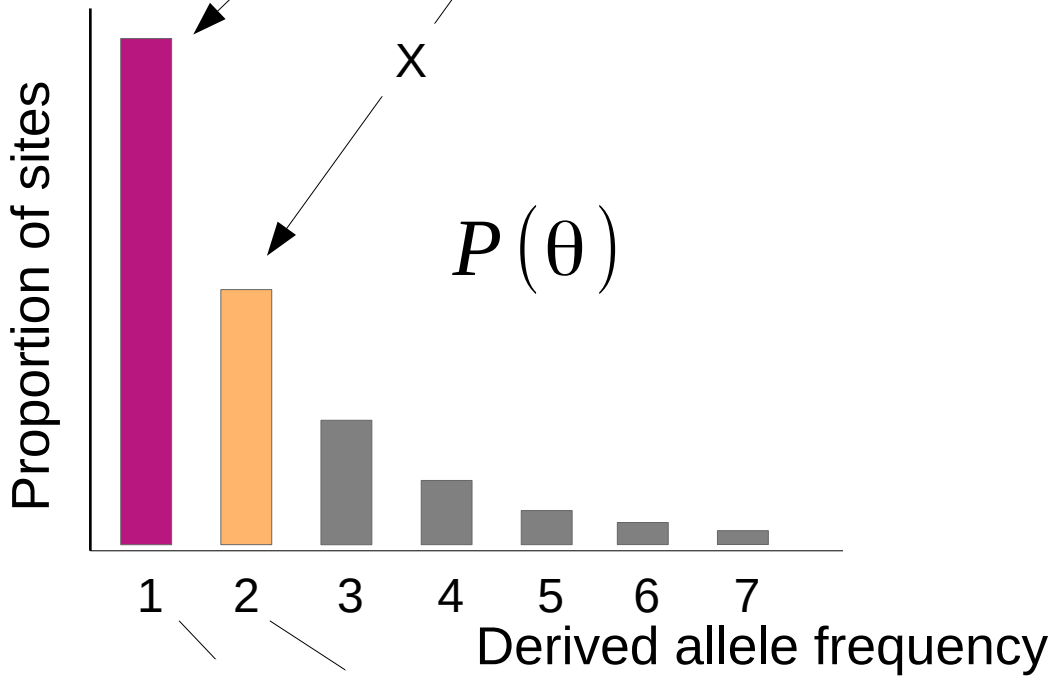
$$P(\theta|X) = P(X|\theta)\,P(\theta)$$

	0	1	2	3	4	.	.	.	2*n*
site1	0.00	-2.24	-4.53	-6.99	-9.63				-232.69
site2	0.00	-2.24	-4.53	-6.99	-9.63				-232.69

$P(X|\theta)$

X

X



=

=

	0	1	2	3	4	.	.	.	2*n*
site1	0.9892	0.0101	0.0006	0.0000	0.0000				0.0000
site2	0.9892	0.0101	0.0006	0.0000	0.0000				0.0000

$\boldsymbol{P(\theta|X)}$

!!! Note that with -doSaf 2 (accounts for inbreeding) the values in the .saf file are already probabilities (not likelihoods) so to get the expected genome-wide SFS sum the columns.

	0	**1**	**2**	**3**	**4**	**5**	**. . .**	***2n***
site1								
site2								
site3								
site4 **.** **.** **.** **.** **.** **.**								
site*m*								

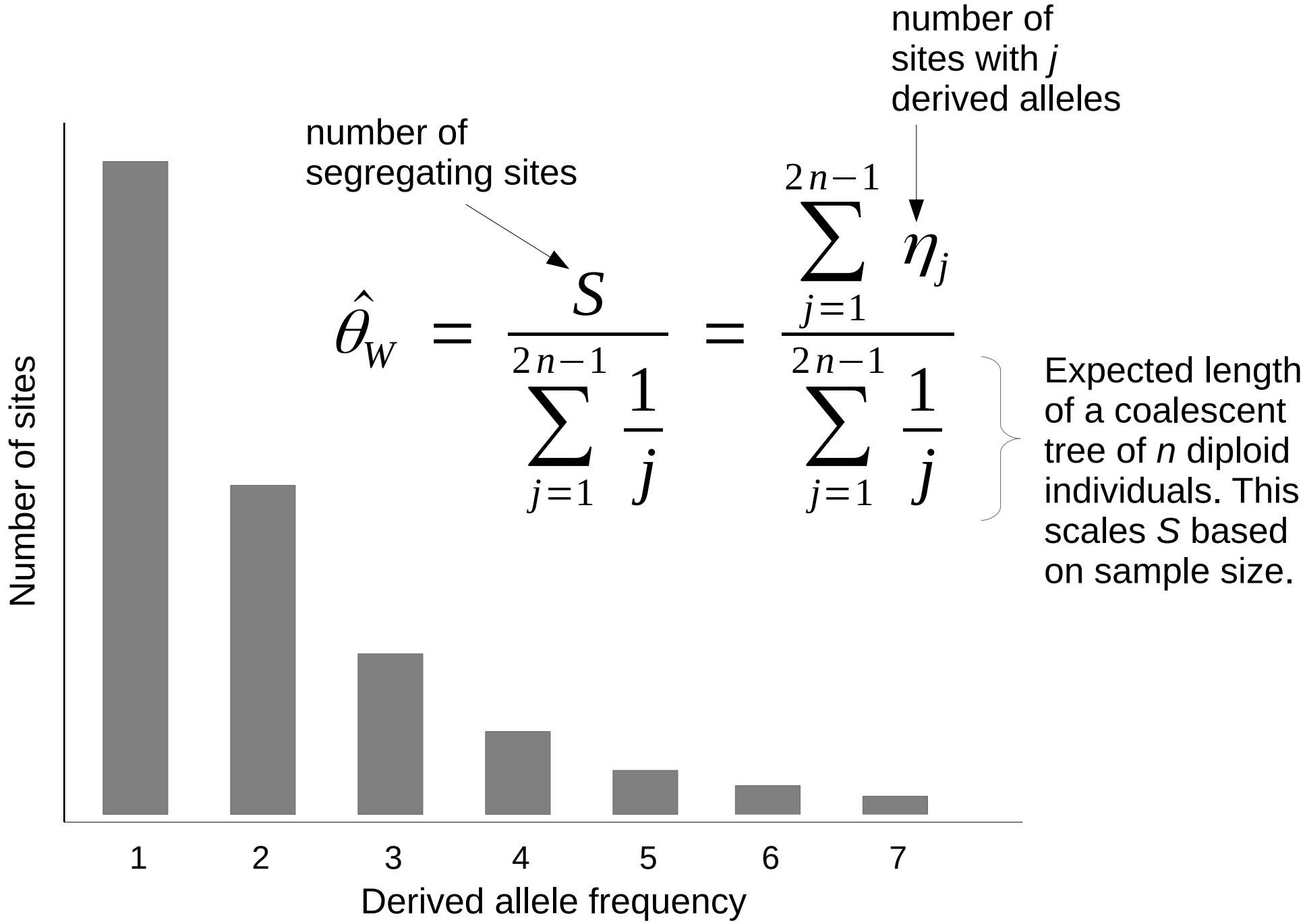
← # derived alleles

column sum → $= p_1$

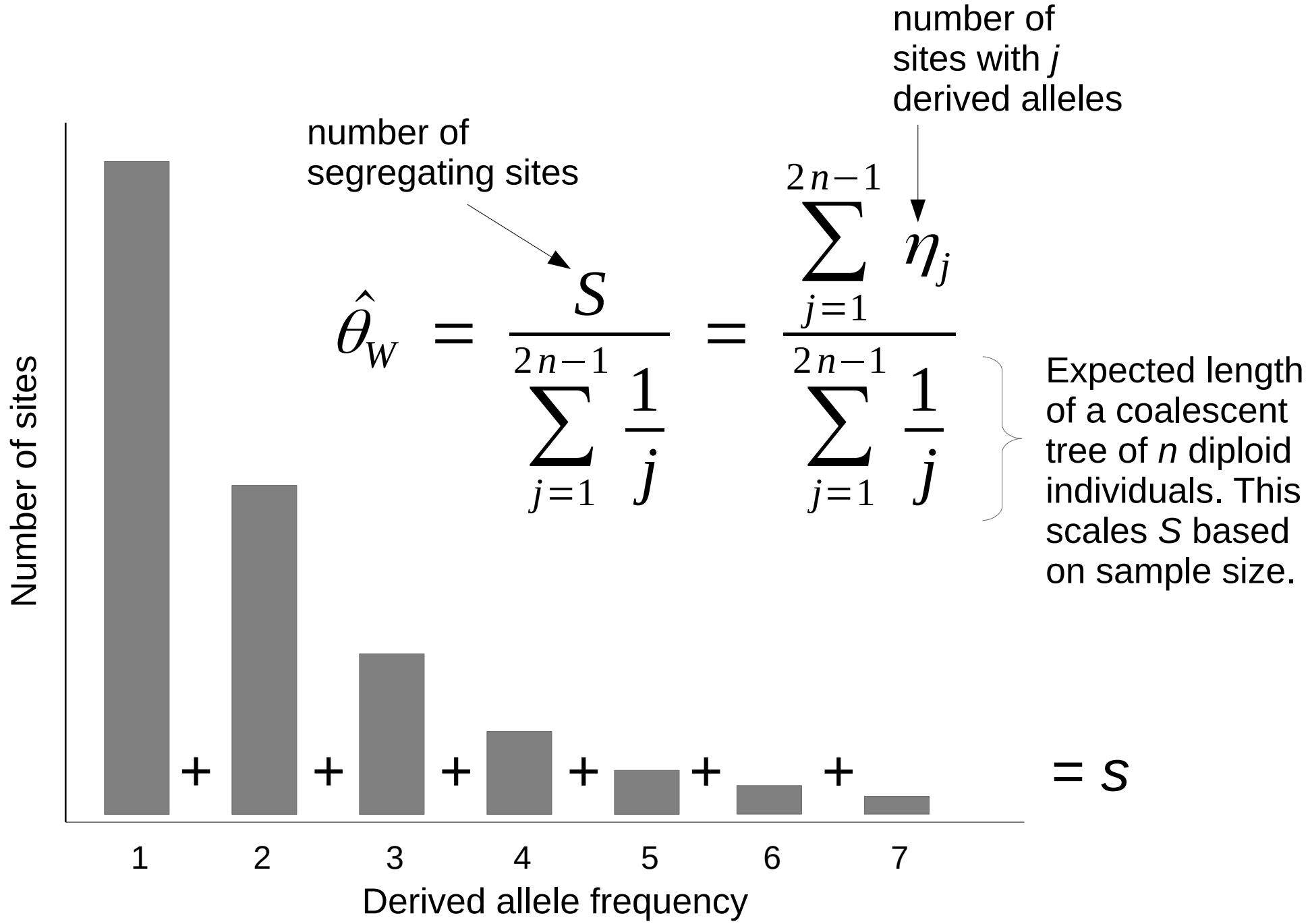
column sum → $= p_4$ . . . . . .

Population genetic summary statistics derived from the SFS

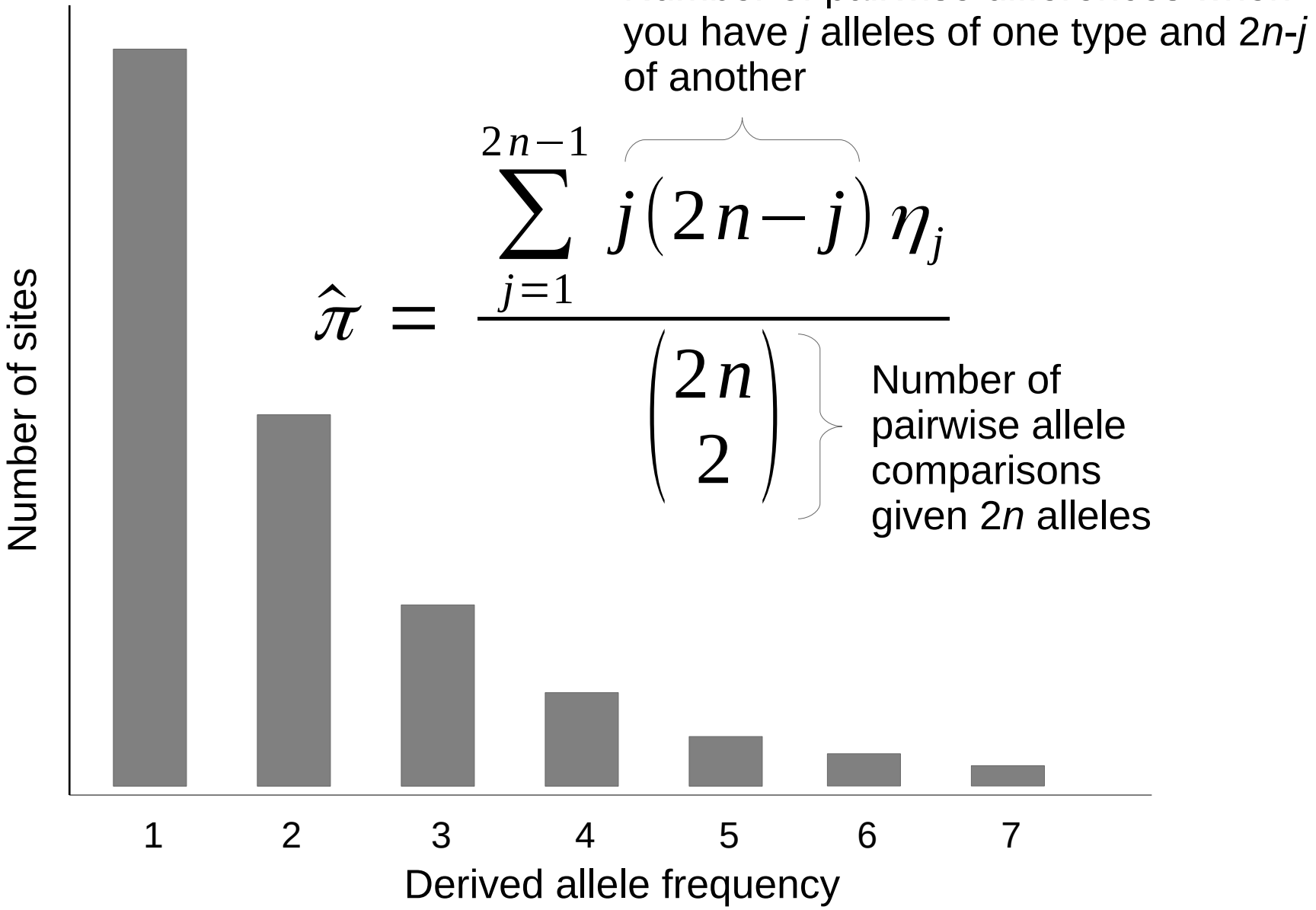
Estimating diversity from the SFS



Estimating diversity from the SFS

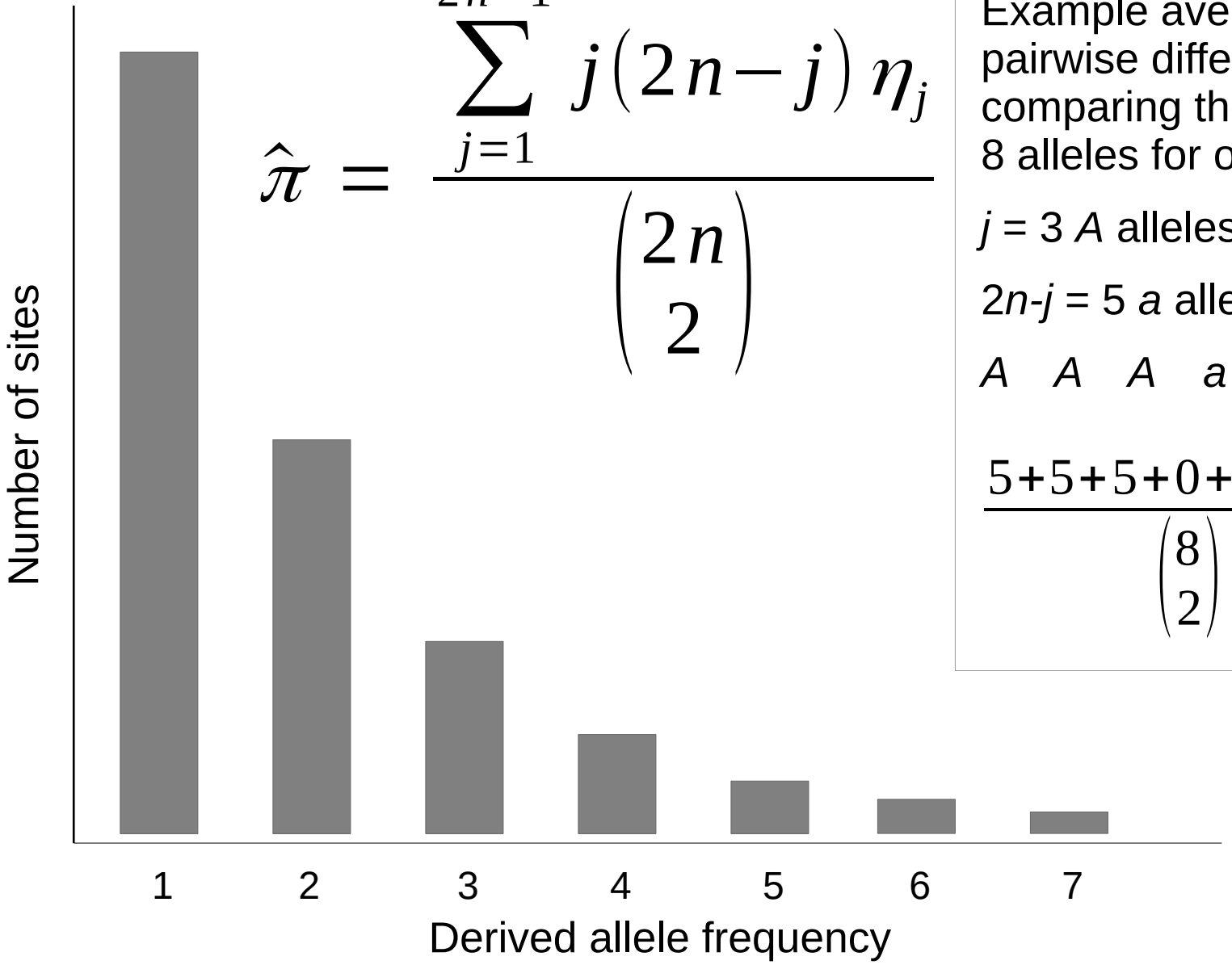


Estimating diversity from the SFS



Estimating diversity from the SFS

$$\hat{\pi} = \frac{\sum_{j=1}^{2n-1} j(2n-j)\eta_j}{\binom{2n}{2}}$$



Example average number of pairwise differences comparing the following set of 8 alleles for one site:

j = 3 *A* alleles

2*n*-*j* = 5 *a* alleles:

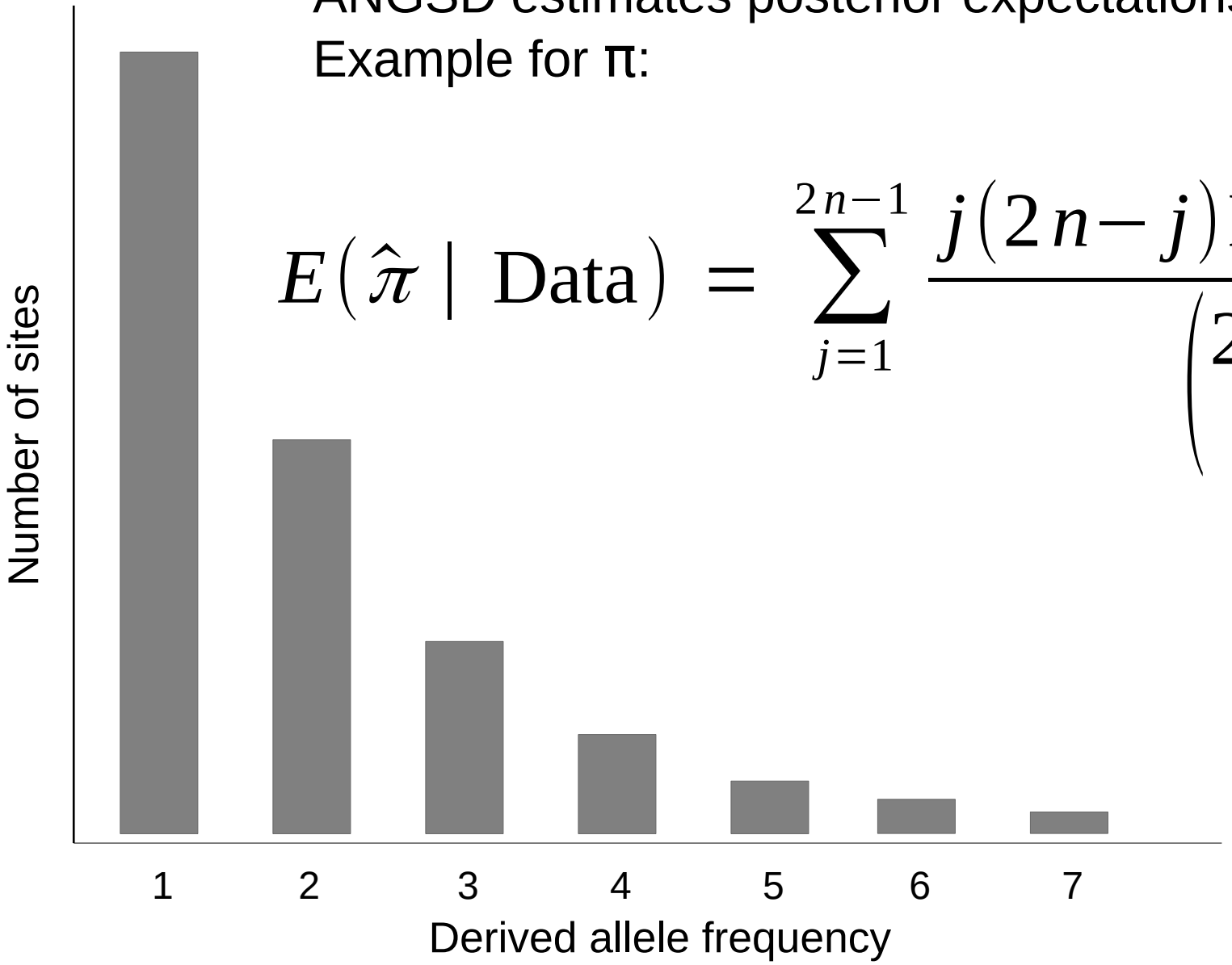
A *A* *A* *a* *a* *a* *a* *a*

$$\frac{5+5+5+0+0+0+0}{\binom{8}{2}} = \frac{15}{28}$$

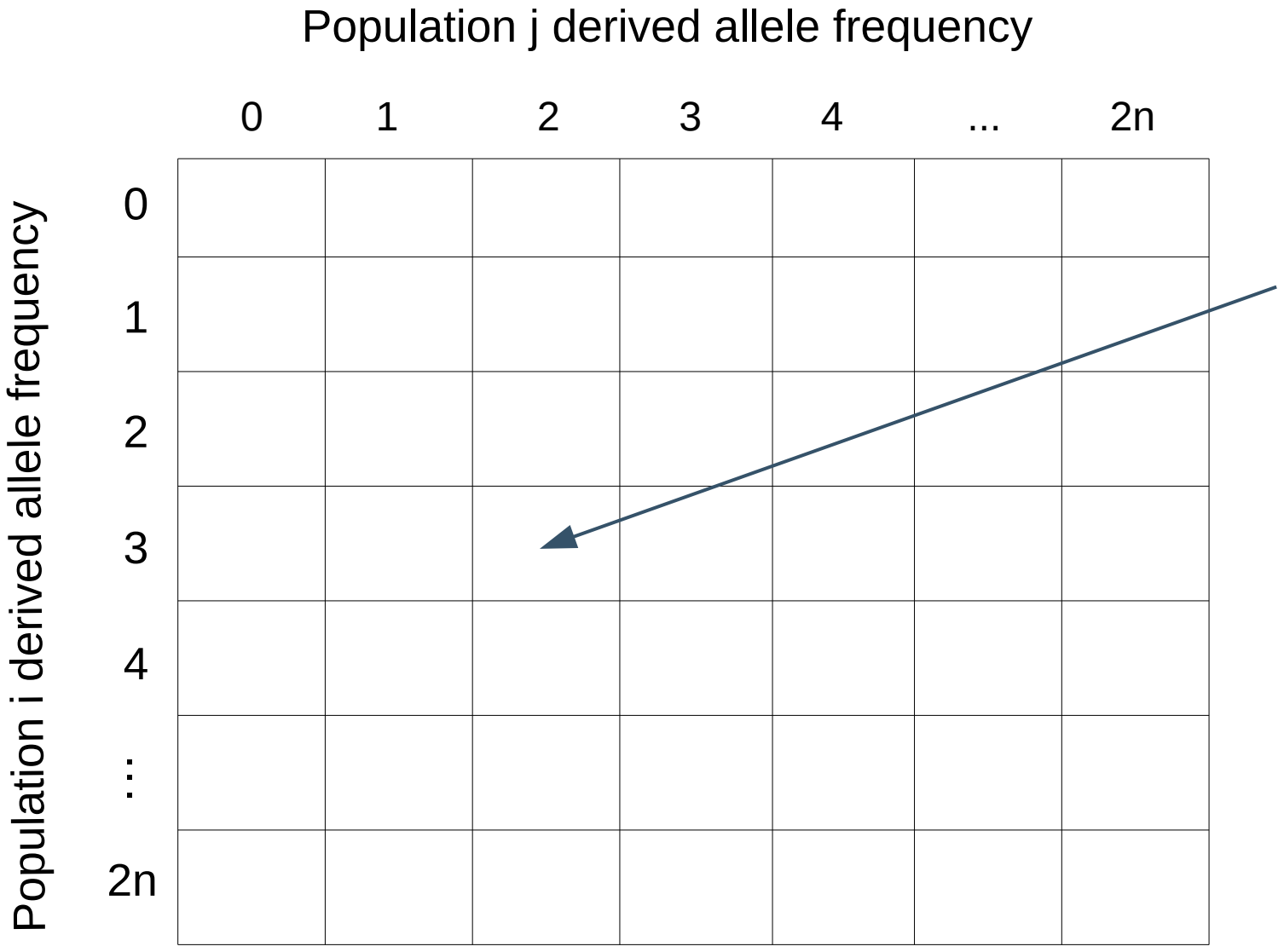
Estimating diversity from the SFS

ANGSD estimates posterior expectations of θ.
Example for π:

$$E\left(\hat{\pi} \mid \text{Data}\right) = \sum_{j=1}^{2n-1} \frac{j\left(2n-j\right)\mathrm{E}\left(\eta_j \middle| \text{Data}\right)}{\binom{2n}{2}}$$



2-dimensional SFS



Proportion of sites where:

pop *i* has 3 derived alleles

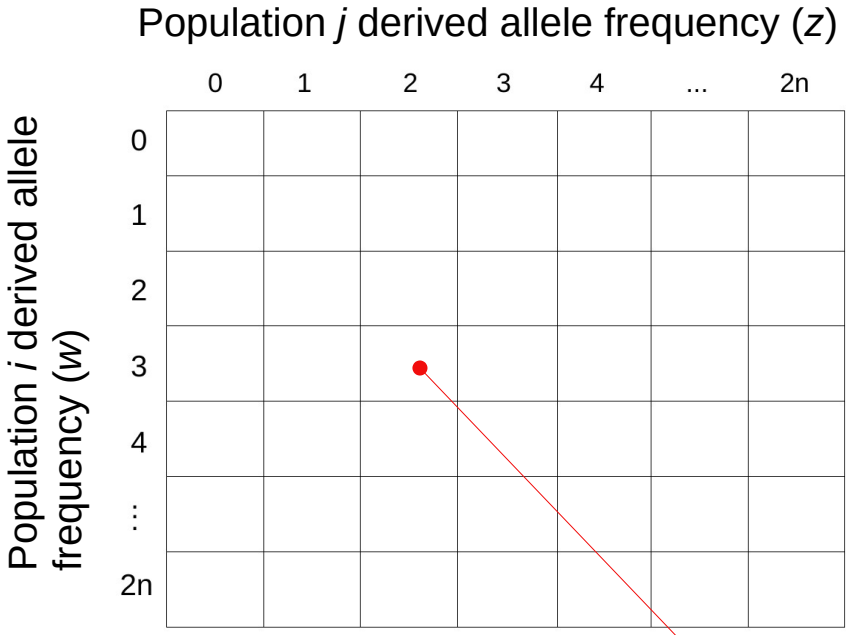
&

pop *j* has 2 derived alleles.

$$F_{ST} = \frac{E(a|X)}{E(a|X)+E(b|X)}$$

Genetic variance between populations

Genetic variance within populations



Probability of *k* derived alleles in pop *i* and *z* derived alleles in pop *j*

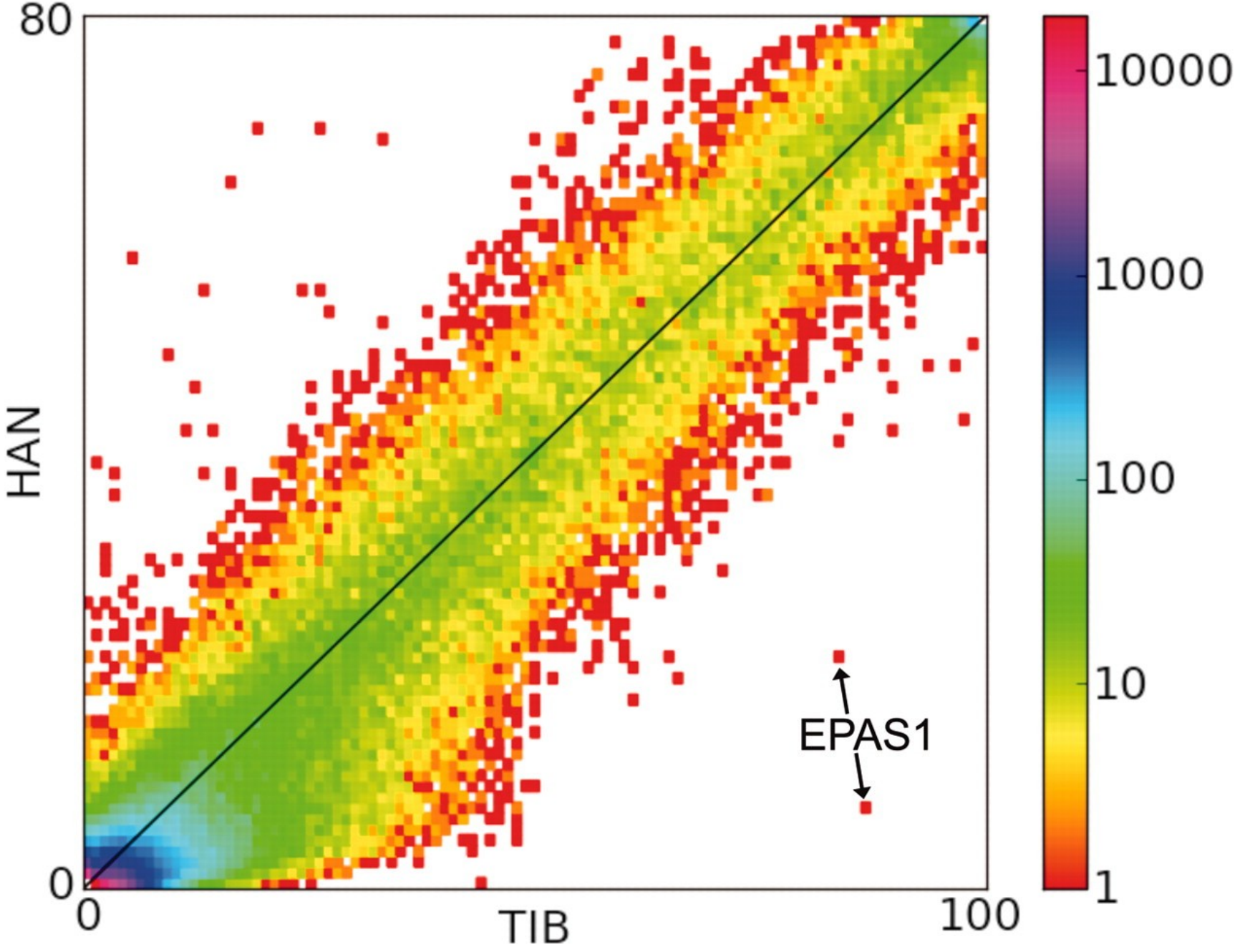
Likelihood of *w* derived alleles in pop *i* at site *v*

Likelihood of *z* derived alleles in pop *j* at site *v*

$$\mathrm{E}(a|X) = \sum_{k=0}^{2n} \sum_{z=0}^{2n} a^{w,z}_{pop\,i,\,pop\,j} \mathrm{P}(X_{i,v}|K_{i,v}=w) \mathrm{P}(X_{j,v}|K_{j,v}=z) Q^{w,z}_{i,j}$$

$$\mathrm{E}(b|X) = \sum_{k=0}^{2n} \sum_{z=0}^{2n} b^{w,z}_{pop\,i,\,pop\,j} \mathrm{P}(X_{i,v}|K_{i,v}=k) \mathrm{P}(X_{j,v}|K_{j,v}=z) Q^{w,z}_{i,j}$$

Identifying loci under selection



Yi *et al*. (2010)

Exercise. Estimating the SFS and related summary statistics